Thermodynamics

It explains the relation b/w "Heat & mechanical coork".

System & Surroundings:

* System is the region of space that contains collection of molecules with in the real or imaginary boundary.
* when the things that present around the system which can exchange either energy, matter or both called surroundings.

Types of system:

① open system: can exchange both energy & matter with it's surrondings called open system.

② Closed system: * pressure of gas is constant.
System can exchange only energy but not matter with its surronding is called closed system.

* Volume of gas is constant *

③ Isolated system: can exchange neither heat nor matter with it's surrounding is called isolated system.

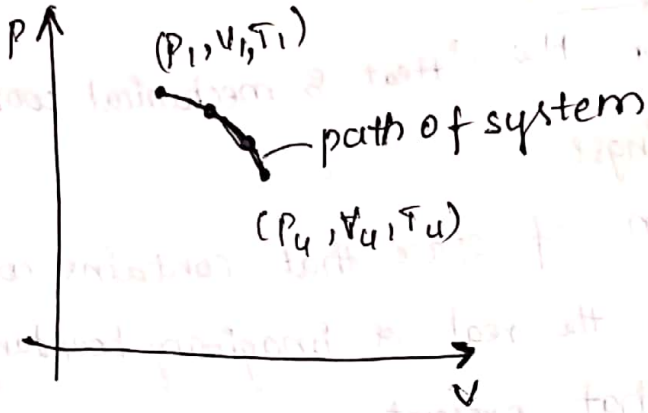
Thermodynamic Variables:

These variables are used

Eqn of States:

It represents the relation b/w thermodynamic variables of the system at a particular state is called eqn of state.

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Internal energy (U) of the system

└ due to K.E — due to their motion

P.E — due to intermolecular forces of attraction b/w the molecules.

$$\boxed{U = \frac{f}{2} nRT}$$

Clearly $U \propto T$

As $T \uparrow \Rightarrow U \uparrow$

$T \downarrow \Rightarrow U \downarrow$

$T = 0 \Rightarrow \boxed{U = 0}$

* Internal energy is a state dependent func but doesnot depends on path followed b/w initial & final states.

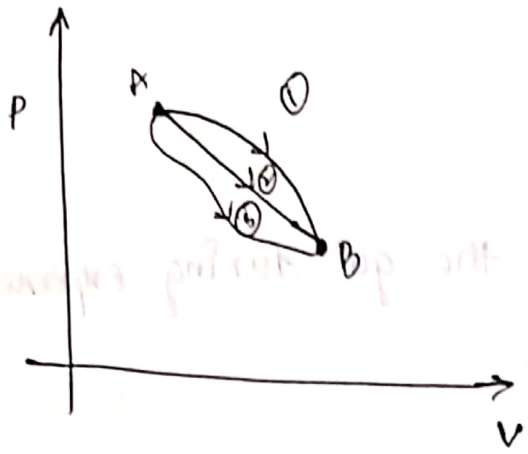
* change in internal energy,

$$\Delta U = U_f - U_i$$

$$\boxed{\Delta U = \frac{f}{2} nR(T_f - T_i)}$$

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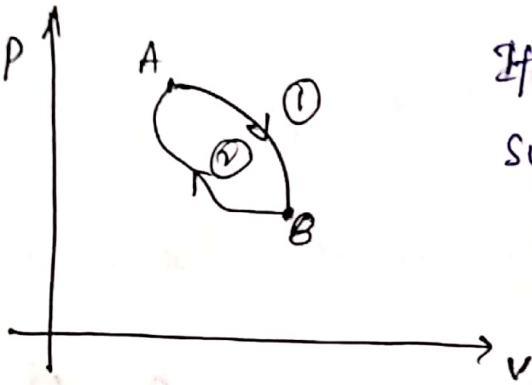


Here $\Delta U = U_f - U_i$

Clearly,

$$\boxed{\Delta U_1 = \Delta U_2 = \Delta U_3}$$

*



If initial & final states of system are same then it is called "cyclic process".

For path 1,

$$\Delta U_1 = U_B - U_A$$

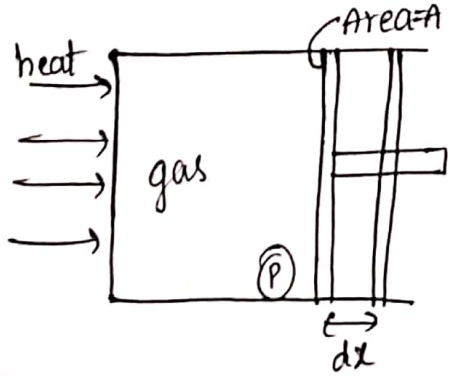
for path 2,

$$\Delta U_2 = U_A - U_B$$

∴ Total change in internal energy of system in cyclic process $(A \rightarrow B \rightarrow A) \Rightarrow \Delta U = \Delta U_1 + \Delta U_2$

$$\boxed{\Delta U = 0}$$

External work done by gas :



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Force on the piston;

$$\boxed{F = PA}$$

small work done by the gas during expansion

$$dw = F\,dx$$

$$dw = PA\,dx$$

$$\boxed{dw = P\,dV} \quad **$$

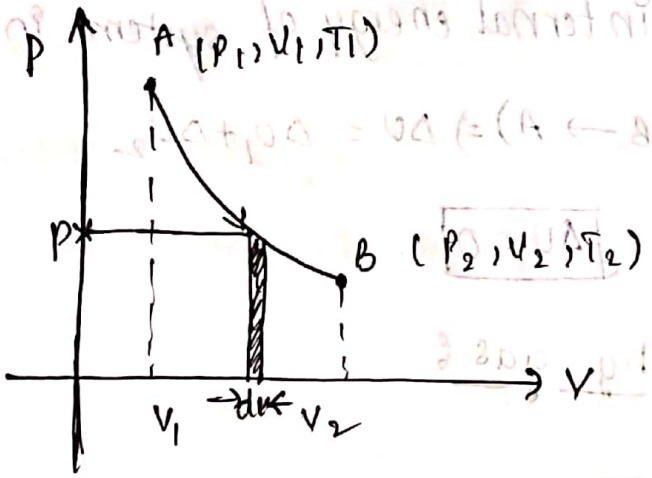
Total work done

$$w = \int dw = \int_{V_1}^{V_2} P\,dV = \text{area of region b/w the curve \& volume axis.}$$

If $P = \text{const}$

$$w = P\int_{V_1}^{V_2} dV \Rightarrow \boxed{w = P(V_2 - V_1)}$$

*



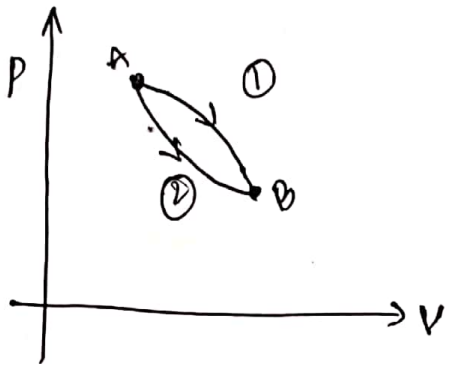
$$dw = P\,dV$$

$$w = \int_{V_1}^{V_2} P\,dV$$



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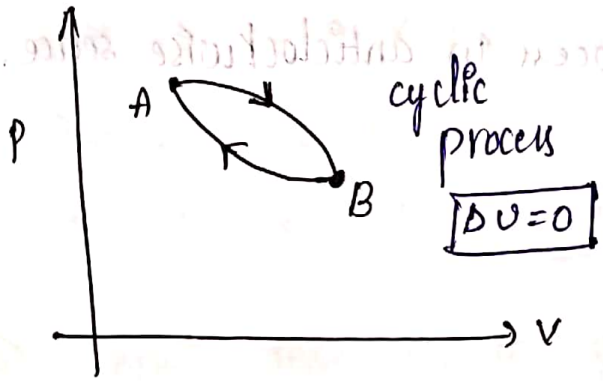
Here $\Delta V_1 = \Delta V_2$

$W_1 = +ve$

$W_2 = +ve$ & $W_1 > W_2$

i.e, work done is a "path dependent quantity"

*



cyclic process

$\Delta U = 0$

Here, $W_1 = +ve$

$W_2 = -ve$

$|W_1| > |W_2|$

* w.done on the gas → +ve
* w.done by the gas → -ve

net work done $W \Rightarrow W_1 + W_2 = +ve$

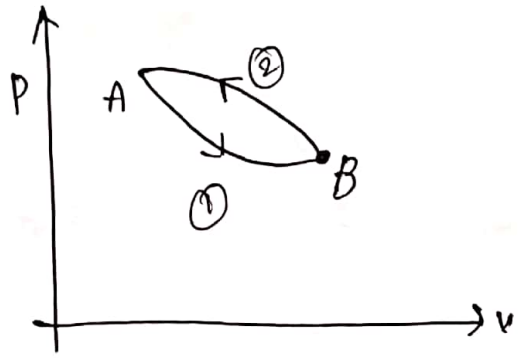
Cyclic process is in clockwise sence.

W = area of the closed loop

** only for P-V diagram.

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$w_1 = +ve$

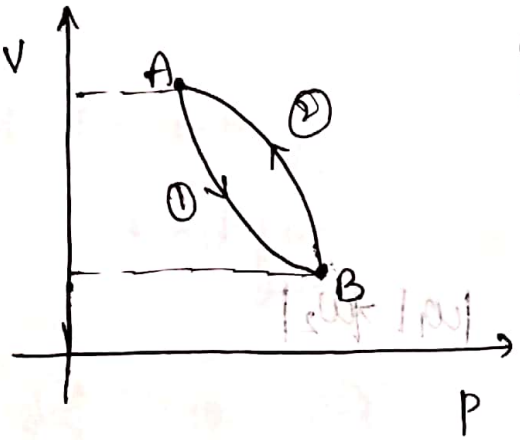
$w_2 = -ve$ & $|w_1| < |w_2|$

net work done,

$$w = w_1 + w_2 = *-ve$$

For cyclic process in anticlockwise sence.

*



$w_1 = -ve$

$w_2 = +ve$ & $|w_2| > |w_1|$

Net work done,

$$w = w_1 + w_2 = +ve$$

First Law of thermodynamics:

A part of heat is utilised by change in internal energy and to some external work.

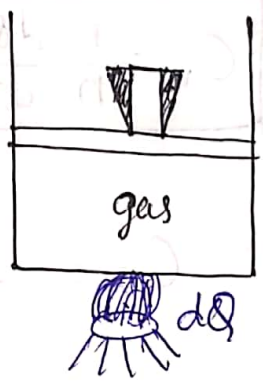
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$$\Delta Q = \Delta U + \Delta W$$

$$Q = (U_f - U_i) + W$$

cal J erg J

$1\ cal = 4.186\ J$ $\quad 1\ J = 10^7\ erg$.



Sign convention:

(1) Heat supplied to the system is taken as +ve.

i.e, $\boxed{\Delta Q = +ve}$

(2) Heat rejected from the system is taken as -ve.

i.e, $\boxed{\Delta Q = -ve}$

(3) work done by the gas during expansion is +ve. i.e, $\boxed{\Delta W = +ve}$

(4) work done by the gas during compression is -ve. i.e, $\boxed{\Delta W = -ve}$

finally,

$$\Delta U = \Delta Q - \Delta W$$

If $\Delta U = +ve \Rightarrow U_f > U_i$ & $\boxed{\Delta T = +ve}$

$\Delta U = -ve \Rightarrow U_f < U_i$ & $\boxed{\Delta T = -ve}$

Specific heats of a gas:

(a) Specific heat of a gas at const volume,

$$\boxed{C_v = \frac{1}{m}\frac{dQ_v}{dT}} \longrightarrow J\,kg^{-1}\,K^{-1}.$$

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& $\boxed{C_v = \frac{1}{n}\frac{dQ_v}{dT}} \rightarrow J\,mol^{-1}K^{-1}$

here, $C_v = \frac{M}{m}\frac{dQ_v}{dT}$

$$\boxed{C_v = Mc_v}$$

(b) Specific heat of gas at constant pressure,

$$c_p = \frac{1}{m}\frac{dQ_p}{dT} \longrightarrow J\,kg^{-1}K^{-1}$$

& $C_p = \frac{1}{n}\frac{dQ_p}{dT} \longrightarrow J\,kg^{-1}K^{-1}$

$\Rightarrow$ $\boxed{C_p = Mc_p}$

To show that $C_p > C_v$:-

At const volume,

$dV = 0$

$pdv = 0$

$\boxed{dw = 0}$

from 1st law of thermodynamics,

$dQ_v = dU + dw$ (dw → 0)

$\boxed{dQ_v = dU}$

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$$\boxed{dU = nC_V\,dT_1}$$

& $$\boxed{dU = \frac{f}{2} nR\,dT_1}$$

$$\boxed{C_V = \frac{fR}{2}}$$

At const pressure :

$$dQ_p = dU + dW$$

$$nC_p\,dT_2 = nC_V\,dT_1 + dW$$

for the same raise in temp in both the cases, i.e, $dT_1 = dT_2$.

$$nC_p\,dT = nC_V\,dT + dW$$

$$C_p = C_V + \frac{dW}{n\,dT}$$

clearly, $\boxed{C_p > C_V}$

$$\Rightarrow \boxed{dQ_p > dQ_V}$$

In general,

$$\boxed{\gamma = \frac{C_p}{C_V}}$$

adiabatic constant.

To prove that $C_p - C_V = R$:-

① At const volume,

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$$dQ_v = dU + \cancel{dW}^{\to 0}$$

$$dQ_v = dU$$

$$nC_v dT = dU \quad -(1)$$

(2) at const pressure,

$$dQ_p = dU + dW$$

$$nC_p dT = nC_v dT + P dV \quad -(2)$$

from Ideal gas eqn,

$$PV = nRT$$

$$P dV + V\cancel{dP}^{\to 0} = nR dT$$

$$\boxed{P dV = nR dT}$$

sub in eqn (2)

$$\cancel{n}C_p \cancel{dT} = \cancel{n}C_v \cancel{dT} + \cancel{n}R\cancel{dT}$$

$$\boxed{C_p - C_v = R}$$ **

for any no. of moles.

constant for any gas.

$$MC_p - MC_v = R$$

$$\boxed{C_p - C_v = R/M} \Rightarrow \boxed{C_p - C_v = s}$$

diff for diff gases.

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Different types of Thermodynamic Processes:

① Isochoric Process:

In this process,

$$V = \text{const.}$$

$$dv = 0$$

$$\Rightarrow dw = pdv = 0$$

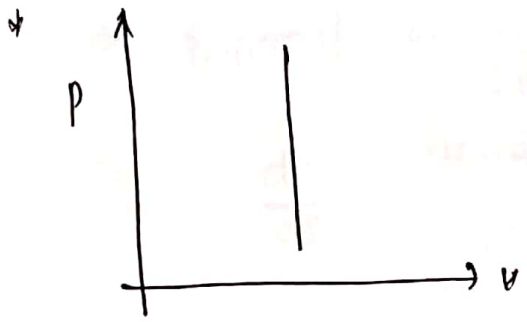
from 1st law of thermodynamics,

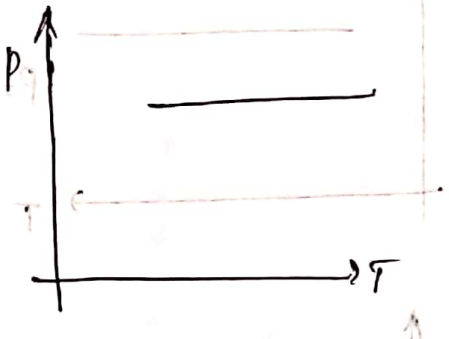
$$dQ_v = dU + \cancel{dw}^{0}$$

$$\boxed{dU = nC_v dT}$$

∴ Specific heat of gas in isochoric process is,

$$\boxed{C_v = \frac{1}{n}\frac{dQ_v}{dT}}$$





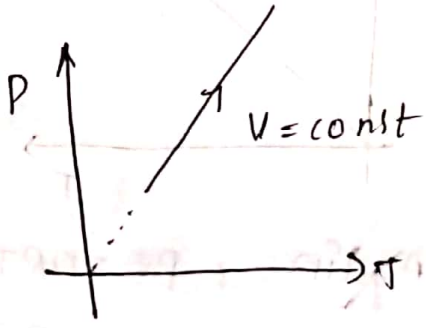
From Ideal gas eqn,

$$PV = nRT$$

$$P = \frac{nRT}{V}$$

$$P \propto T$$

$$\boxed{\frac{P_1}{P_2} = \frac{T_1}{T_2}} \quad \text{— Charle's law}$$



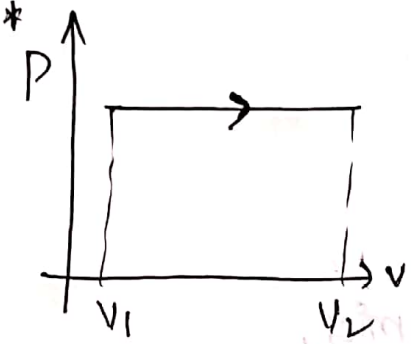
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② Isobaric Process :

In this process,

$P = \text{const.}$

*



$$W = P(V_2 - V_1)$$

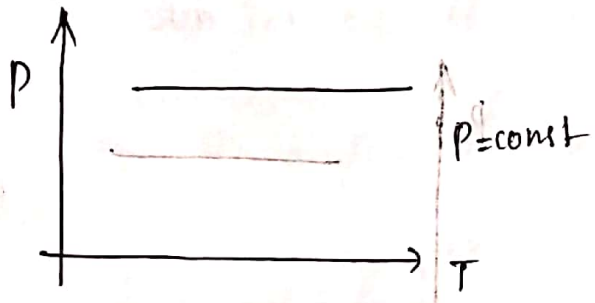
$$PV = nRT$$

$$PV_1 = nRT_1$$

$$PV_2 = nRT_2$$

$$\Rightarrow \boxed{W = nR(T_2 - T_1)}$$

*



*



Again, $PV = nRT$

$$V = \frac{nRT}{P}$$

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$V \propto T$

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

Charle's law.

* Specific heat of gas in isobaric process $= C_p$.

we have,

$$C_p = \frac{1}{n} \frac{dQ_p}{dT}$$

$\therefore$ heat supplied in isobaric process,

$$dQ_p = nC_p dT$$

From 1st law of Thermodynamics,

$$dQ_p = dU + dW$$

* fraction of heat is utilized in increasing the internal energy,

$$\frac{dU}{dQ_p} = \frac{nC_v dT}{nC_p dT}$$

$$\frac{dU}{dQ_p} = \frac{C_v}{C_p} = \frac{1}{\gamma}$$

&

$$\frac{dU}{dQ_p} \times 100 = \frac{1}{\gamma} \times 100\ \%$$

* fraction of heat is utilized in doing the external work,

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$$\frac{dw}{dQ_p} = \frac{dQ_p - dU}{dQ_p}$$

$$= 1 - \frac{dU}{dp}$$

$$= 1 - \frac{c_v}{c_p}$$

$$= 1 - \frac{1}{\gamma}$$

$$= \frac{\gamma - 1}{\gamma}$$

$$\% \quad \boxed{\frac{dw}{dQ_p} \times 100 = \frac{\gamma - 1}{\gamma} \times 100\%}$$

Isothermal process:

* It is the process in which physical changes in the system occurs at const temp.

Here $T = \text{const}$

$$\Rightarrow \boxed{dT = 0}$$

$$\Rightarrow \boxed{dU = 0}$$

From 1st law of thermodynamics,

$$dQ = \cancel{dU}^{0} + dw$$

$$dQ = dw$$

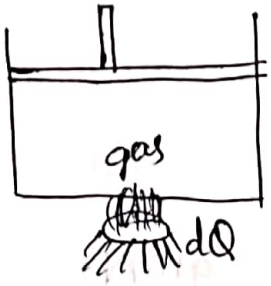
$$\text{or } \boxed{Q = w}$$

i.e, quantity of heat supplied = amount of

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work done by system.

Condns:



① walls of container & piston are made with good conductors of heat.

② process is very slow.

③ we have, $PV = nRT$.
(nRT : const)

$$\boxed{PV = \text{const}}$$

i.e, gas obeys Boyle's law.

And, $PV = \frac{m}{M}RT$

$$P = \frac{\rho RT}{M}$$

$$\boxed{P \propto \rho}$$

Graphs:

① $PV = \text{const}$



Slope of curve,

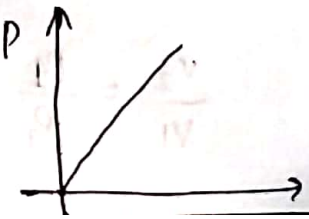
$P dv + v dp = 0$

$$\boxed{\frac{dP}{dv} = -\frac{P}{V}}$$

Isothermal Bulk modulus of gas, $B_{iso} = \frac{-dP}{\left(\frac{dV}{V}\right)_0} = P$

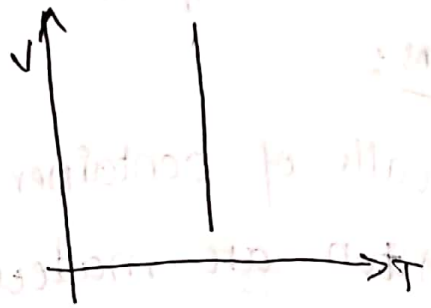
slope of isothermal curve is "negative".

② P vs ρ graph

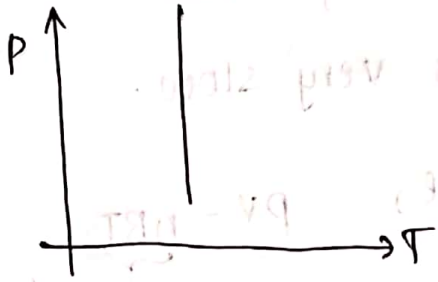


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③ V vs T graph



④ P vs T graph



Work done by gas in isothermal process:

small amount of work done,

$$dw = P\,dV$$

Total work done,

$$w = \int dw = \int_{V_1}^{V_2} P\,dV$$

$$w = \int_{V_1}^{V_2} \frac{nRT}{V}\,dV$$

$$* \boxed{w = nRT\,ln(V_2/V_1)} *$$

$$w = nRT(2.303)\log(V_2/V_1)$$

$$\boxed{w = 2.303nRT\log(V_2/V_1)}$$

we have,

$$P_1V_1 = P_2V_2$$

$$\frac{V_2}{V_1} = \frac{P_1}{P_2}$$

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$$\boxed{w = 2.303\, nRT \log\left(\frac{P_1}{P_2}\right) = Q}$$

Adiabatic Process

* Specific heat of gas in isothermal process,

$$C = \frac{1}{n}\frac{dQ}{dT}$$

here $dT = 0$

$$\Rightarrow \boxed{C = \infty}$$

Adiabatic process:

It is the process in which physical changes in the system occurs without exchange of heat with system & surrondings.

Here $\boxed{dQ = 0}$

from 1st law of thermodynamics,

$$dQ = dU + dW$$

$$dU + dW = 0$$

$$\boxed{dU = -dW}$$

If $dW = +ve \Rightarrow \boxed{dU = -ve}$

$$\Rightarrow \boxed{dT = -ve}$$

i.e, temp of system decreases during adiabatic expansion of gas.

If $dW = -ve \Rightarrow \boxed{dU = +ve}$

$$\Rightarrow \boxed{dT = +ve}$$

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i.e, temp increases during adiabatic compression of gas.

Condns:



① container walls & piston are made with good insulator of heat

② process is quick.

③ Here gas obey "poisson's law"

$$PV^{\gamma} = \text{const}$$

$\gamma = \frac{C_P}{C_V}$ = adiabatic const.

We have,

$PV = nRT$

$\frac{nRT}{V} \times V^{\gamma} = \text{const}$

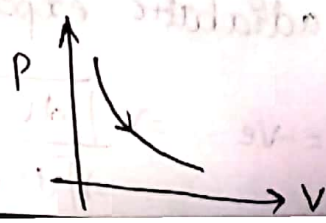
$$TV^{\gamma-1} = \text{const}$$

and $P\left(\frac{nRT}{P}\right)^{\gamma} = \text{const}$

$$P^{1-\gamma} \cdot T^{\gamma} = \text{const}$$

Graphs:

① P-V graph:



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slope of the curve

$$PV^{\gamma} = \text{const}$$

$$P\gamma V^{\gamma-1} dV + V^{\gamma} dP = 0$$

$$\boxed{\frac{dP}{dV} = -\frac{\gamma P}{V}}$$

└ slope = -ve

* Adiabatic Bulk modulus of gas

$$B_{adia} = \frac{-dP}{\frac{dV}{V}}$$

$$\boxed{B_{adia} = \gamma P}$$

Imp pts:

$$\frac{(\text{slope})_{adi}}{(\text{slope})_{iso}} = \frac{+\gamma P}{+P}$$

$$\boxed{(\text{slope})_{adia} = \gamma\,(\text{slope})_{iso}}$$

i.e, adiabatic curve is more steeper than isothermal curve

② T-V graph:

$$TV^{\gamma-1} = \text{const}$$

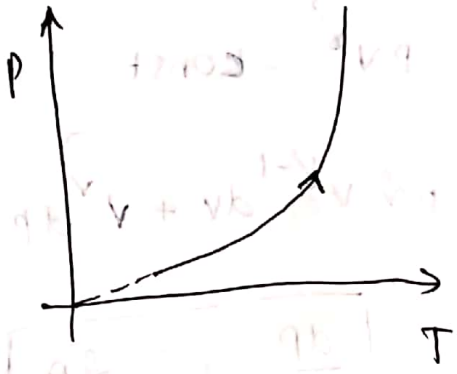
as $T \uparrow \Rightarrow V \downarrow$

* yx^n = const — hyperbola



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③ P–T graph:



$$P^{1-\gamma} \cdot T^{\gamma} = \text{const}$$

$$PT^{\gamma/1-\gamma} = \text{const}$$

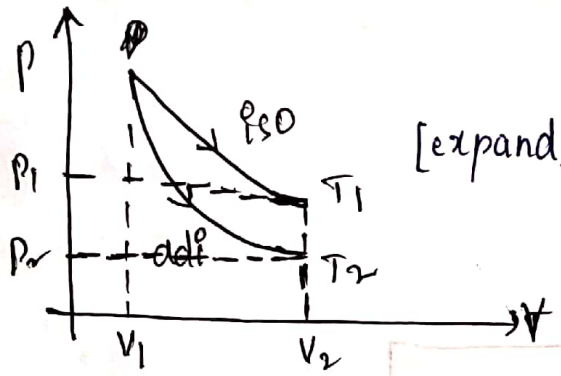
$$PT^{-\gamma/\gamma-1} = \text{const}$$

$$\frac{P}{T^{\frac{\gamma}{\gamma-1}}} = \text{const}$$

$$\boxed{P = \text{const} \times T^{\frac{\gamma}{\gamma-1}}}$$

as $P\uparrow \Rightarrow T\uparrow$

Points:



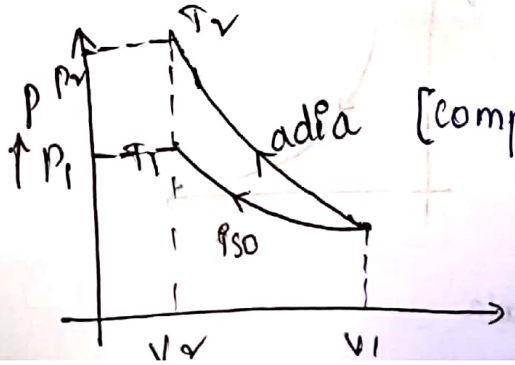
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① $P_1 > P_2$

② $T_1 > T_2$

③ $W_{iso} > W_{adia}$

*



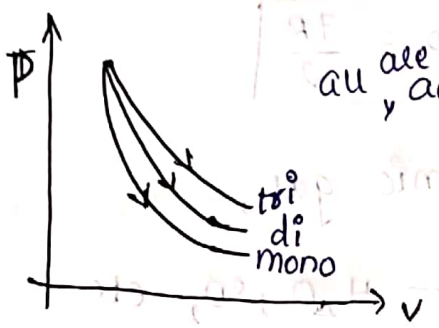
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① $P_1 < P_2$

② $T_1 < T_2$

③ $W_{adia} > W_{iso}$

* 

all are adiabatic curves.

$\text{slope} = \gamma P$

more γ, more steeper.

Relation b/w γ & f :

we have,

① $\gamma = 1 + \frac{2}{f}$

② $C_v = \frac{fR}{2}$, $C_p = \left(\frac{f+2}{2}\right)R$

③ $\gamma = \frac{C_p}{C_v}$ & $C_p - C_v = R$

$\Rightarrow \boxed{C_v = \frac{R}{\gamma - 1}}$ & $\boxed{C_p = \frac{\gamma R}{\gamma - 1}}$

* For monoatomic gas,

└ He, Ne, Ar, Kr, Xe, Ra etc..

$f = 3 \Rightarrow \gamma = 1 + \frac{2}{3}$

$\gamma = \frac{5}{3} = 1.66$

$C_v = \frac{3R}{2}$ & $C_p = \frac{3R}{2} + R = \frac{5R}{2}$

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* For diatomic gas,
  └ H_2, N_2, O_2 etc.. and also air

$f = 5 \Rightarrow \gamma = 1 + \frac{2}{5} = \frac{7}{5} = 1.4$

$$\boxed{C_V = \frac{5R}{2}}, \quad \boxed{C_P = \frac{7R}{2}}$$

* For non-linear triatomic gas,
  └ H_2O, SO_2 etc...

$f = 6 \Rightarrow \gamma = 1 + \frac{2}{6} = \frac{4}{3} = 1.33$

$$\boxed{C_V = 3R}, \quad \boxed{C_P = 4R}$$

Work done by the gas during adiabatic process:

Small work done,

$dW = P\,dV$

we have, $W = \int dW = \int_{V_1}^{V_2} P\,dV$

we have $PV^\gamma = K$

$\Rightarrow P = \frac{K}{V^\gamma}$

$$W = \int_{V_1}^{V_2} \frac{K}{V^\gamma}\,dV$$

$$W = \frac{K}{1-\gamma}\left[V^{-\gamma+1}\right]_{V_1}^{V_2}$$

$$W = \frac{K}{1-\gamma}\left(V_2^{1-\gamma} - V_1^{1-\gamma}\right)$$

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$$W = \frac{KV_2^{1-\gamma} - KV_1^{1-\gamma}}{1-\gamma}$$

$$= \frac{P_2V_2^{\gamma}V_2^{1-\gamma} - P_1V_1^{\gamma}V_1^{1-\gamma}}{1-\gamma}$$

$$\boxed{W = \frac{P_1V_1 - P_2V_2}{\gamma - 1}} \quad *$$

we have,

$$P_1V_1 = nRT_1$$

$$P_2V_2 = nRT_2$$

$$\boxed{W = \frac{nR}{\gamma-1}(T_1 - T_2)} \quad ** *$$

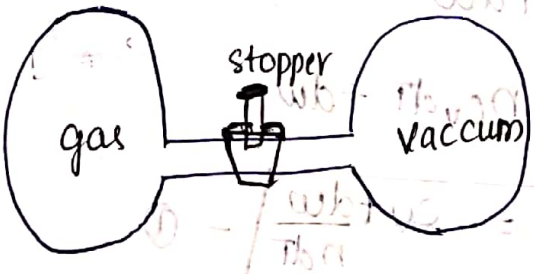
*Specific heat of gas in adiabatic process,

$$C = \frac{1}{n}\frac{dQ}{dT}$$

here $dQ = 0$

$$\Rightarrow \boxed{C = 0}$$

Adiabatic free expansion of gas:



-all are adiabatic walls,

$$dQ = 0$$

when stopper is suddenly opened, gas expands against vaccum.

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Here, $\boxed{dQ = 0}$

from 1st law of Thermodynamics,

$dQ = \cancel{dU}^{0} + dW$ $\Rightarrow \boxed{dU = 0}$

$\Rightarrow \boxed{dT = 0}$

Polytropic process :

Eqn of polytropic process,

$$\boxed{PV^x = \text{const}}$$

where $(x \in (-\infty, \infty))$

① when $x = \pm\infty \Rightarrow V = \text{const}$ (Isochoric process)

② when $x = 0 \Rightarrow P = \text{const}$ (Isobaric process)

③ when $x = 1 \Rightarrow PV = \text{const}$ (Isothermal process)

④ when $x = \gamma \Rightarrow PV^{\gamma} = \text{const}$ (adiabatic process)

* Specific heat of gas in any process,

we have, $dQ = dU + dW$

$$nC_{process}dT = nC_V dT + dW$$

$$\boxed{C_{process} = C_V + \frac{dW}{n\,dT}} \quad -①$$

└ work done per unit mole per unit rise in temp.

$$\boxed{C_{process} = C_V + \frac{\Delta W}{n\,\Delta T}}$$

we have, $PV^x = K$

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$$W = \int P dV = \int_{V_1}^{V_2} \frac{k}{V^x} . dV$$

$$W = \frac{nR}{x-1}(T_1 - T_2) = \frac{nR}{1-x}(T_2 - T_1)$$

$$\frac{W}{n(T_2-T_1)} = \frac{R}{1-x}$$

eqn ① becomes,

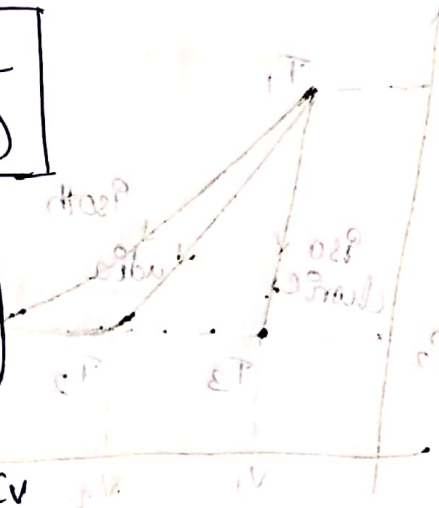
$$C_{process} = C_v + \frac{R}{1-x} \quad **$$

$$\Rightarrow C = \frac{R}{\gamma-1} + \frac{R}{1-x}$$

$$C = R\left(\frac{x - x + \gamma - 1}{(\gamma-1)(1-x)}\right)$$

$$\boxed{C = \frac{R(\gamma - x)}{(\gamma-1)(1-x)}}$$

$$\Rightarrow \boxed{C = C_v \frac{(\gamma - x)}{(1-x)}}$$



when $x = \infty \Rightarrow C = C_v$

$x = 0 \Rightarrow C = C_v + R = C_p$

$x = 1 \Rightarrow C = \infty$

$x = \gamma \Rightarrow C = 0$

① C = +ve when $x < 1$

② C = +ve when $x > \gamma$

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③ C = -ve when $1 < x < \gamma$

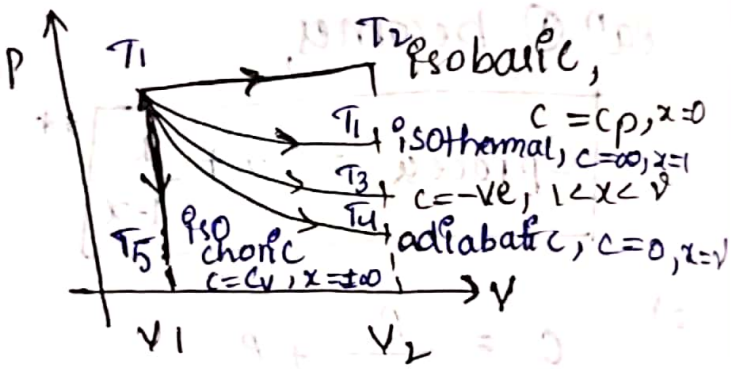
i.e, all the thermodynamic processes lying btw isothermal & adiabatic processes contain -ve specific heat.

Graphs:

① P–V graph:

$PV^x = \text{const}$

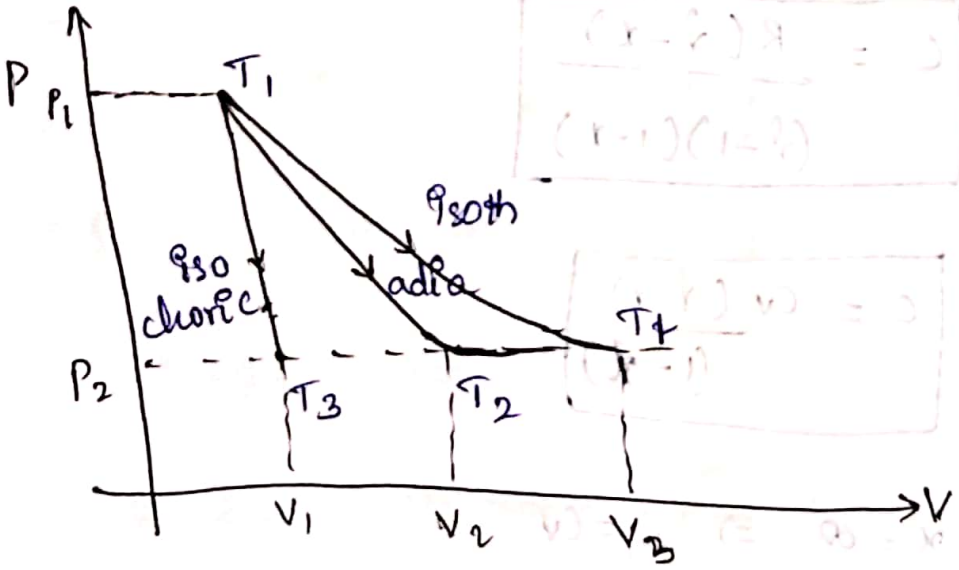
$$C = C_V + \frac{R}{1-x}$$



here $T_2 > T_1 > T_3 > T_4$

& $W_{isob} > W_{isoth} > W_{poly} > W_{adia}$

②



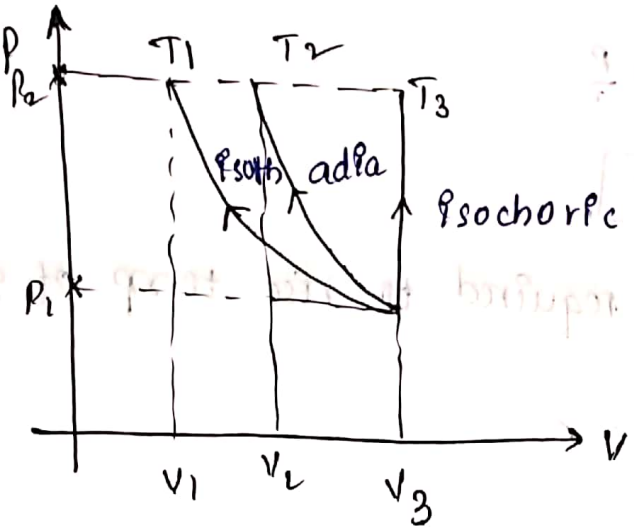
here ① $V_3 > V_2 > V_1$

② $T_1 > T_2 > T_3$

③ $W_{isoth} > W_{adia} > W_{isoch} = 0$

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(3)



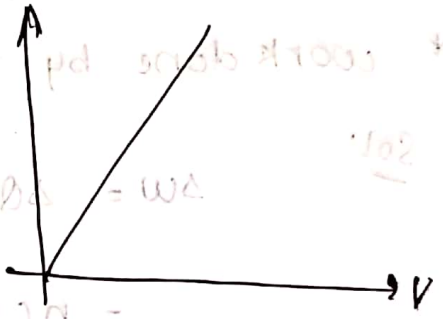
(1) $V_3 > V_2 > V_1$

(2) $T_3 > T_2 > T_1$

(3) $W_{isoth} > W_{adia}$

Eg!

*Find specific heat of a mono-atomic gas in the process as shown.



Sol!

Eqn of st. line, $P = KV$

$PV^{-1} = k$

comparing with $PV^x = k$

$\therefore \boxed{x = -1}$

Specific heat of gas,

$$C = C_V + \frac{R}{1-x}$$

$$C = \frac{3R}{2} + \frac{R}{1-(-1)}$$

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$$C = \frac{3R}{2} + \frac{R}{2}$$

$$\boxed{C = 2R}$$

* Heat energy required to rise temp of gas by dT is

Sol!

$$\Delta Q = nC\Delta T$$

$$\boxed{\Delta Q = n(2R)dT}$$

* change in Internal energy,

Sol:

$$\Delta U = n C_v dT$$

$$= n\left(\frac{3R}{2}\right)dT$$

* work done by the gas,

Sol:

$$\Delta W = \Delta Q - \Delta U$$

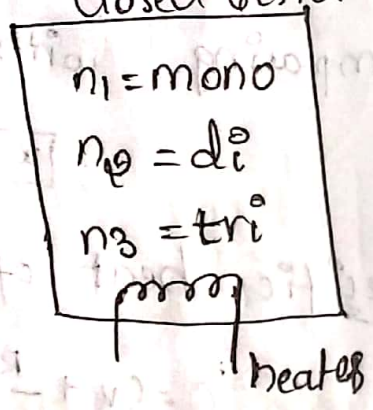
$$= n(2R)dT - n\left(\frac{3R}{2}\right)dT$$

$$= ndT\left(\frac{R}{2}\right)$$

$$\boxed{\Delta W = n\left(\frac{R}{2}\right)\cdot dT}$$

Mixture of gases:

Heat required to rise the temp of mixture by ΔT.



$$\Delta Q = n_1 C_{v_1} \Delta T + n_2 C_{v_2} \Delta T + n_3 C_{v_3} \Delta T$$

$$\Rightarrow \Delta Q = n_1\left(\frac{3R}{2}\right)\Delta T + n_2\left(\frac{5R}{2}\right)\Delta T + n_3\left(\frac{6R}{2}\right)\Delta T$$

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∴ heat capacity,

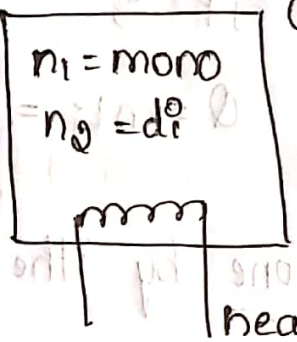
$$\frac{\Delta Q}{\Delta T} = n_1\left(\frac{3R}{2}\right) + n_2\left(\frac{5R}{2}\right) + n_3(3R)$$

* Avg molecular specific heat capacity,

$$C_V = \frac{\frac{\Delta Q}{\Delta T}}{n_1+n_2+n_3} = \frac{n_1\left(\frac{3R}{2}\right) + n_2\left(\frac{5R}{2}\right) + n_3(3R)}{n_1+n_2+n_3}$$

Dissociation of Gases

Let k-fraction of molecule of diatomic gas dissociates into atoms.



then,

no. of moles of mono-atomic gas

$$n_1' = n_1 + 2kn_2$$

no. of moles of diatomic gas

$$n_2' = n_2 - kn_2$$

* Before dissociation,

$$C_V = \frac{n_1\left(\frac{3R}{2}\right) + n_2\left(\frac{5R}{2}\right)}{n_1+n_2}$$

* After dissociation,

$$C_V' = \frac{(n_1+2kn_2)\left(\frac{3R}{2}\right) + (n_2-kn_2)\left(\frac{5R}{2}\right)}{n_1+2kn_2+n_2-kn_2}$$

$$C_V' = \frac{(n_1+2kn_2)\left(\frac{3R}{2}\right) + (n_2-kn_2)\left(\frac{5R}{2}\right)}{n_1+n_2+kn_2}$$

Clearly $C_V' < C_V$ i.e, molecular specific heat

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decreases on dissociation.

Eg: Consider the melting of 1gm of ice at 0°C to liq. water at 0°C at atm pressure. Calculate the workdone on ice system in melting & change in internal energy of system.

Sol:

*Heat supplied,

$$Q = mL_i = 1\,gm \times \frac{80\,cal}{gm} = 80 \times 4.2\,J$$

*work done by the ice,

$$W = P(V_2 - V_1)$$

where $P = 1.013 \times 10^5\ N/m^2$

$$V_1 = \frac{m}{\rho_i} = \frac{1\,gm}{0.92\,gm/cm^3} = \frac{1\,cm^3}{0.92} = \frac{10^{-6}}{0.92}\,m^3$$

& $V_2 = \frac{m}{\rho_w} = \frac{1\,gm}{1\,gm/cm^3} = \frac{1\,cm^3}{1} = 10^{-6}\,m^3$

$$W = 1.013 \times 10^5 \times 10^{-6}\left(1 - \frac{1}{0.92}\right) J$$

$$\boxed{W = -9 \times 10^{-3}\,J}$$

* work done on the ice $= 9 \times 10^{-3}\,J$

Finally, from 1st law of thermodynamics,

$$Q = \Delta U + W$$

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$$\Delta U = Q - W$$

$$= 336 - (-9 \times 10^{-3}) = 336 \text{ J}$$

Ex: Consider the vaporisation of 1gm of water at 100°C to steam at 100°C at 1atm Calculate the work done by the water system in the vapourisation and change in internal energy of the system.

Sol:

* Heat supplied,

$$Q = mL_v$$

$$= 1\text{gm} \times 540 \times \frac{\text{cal}}{\text{gm}}$$

$$= 540 \times 4.2 \text{ J}$$

$$Q = 2268 \text{ J}$$

* work done by the water,

$$W = P(V_g - V_l)$$

$$V_l = \frac{m}{\rho_w} = 1\text{cm}^3 = 10^{-6}\text{m}^3$$

* No. of moles of 1gm of water (H_2O),

$$\eta = \frac{m}{M} = \frac{1\text{gm}}{18\text{gm}} = \frac{1}{18}$$

we have, $PV = nRT$

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$$V_2 = \frac{nRT}{P} = \frac{\frac{1}{18} \times 8.314 \times 273}{1.013 \times 10^5} m^3$$

now,

$$W = P(V_2 - V_1)$$

$$= 172 \text{ J}$$

Vol of 1gm of steam $= 1671 cm^3$

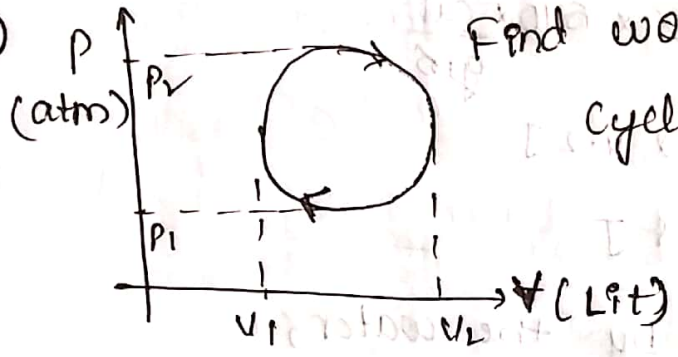
finally,

$$\Delta U = Q - W$$

$$= 2268 - 172$$

$$\boxed{\Delta U = 2.09 \times 10^3 \text{ J}}$$

Graphs:

① 

Find work done by the gas in cyclic process.

Sol:

work done = area of closed loop

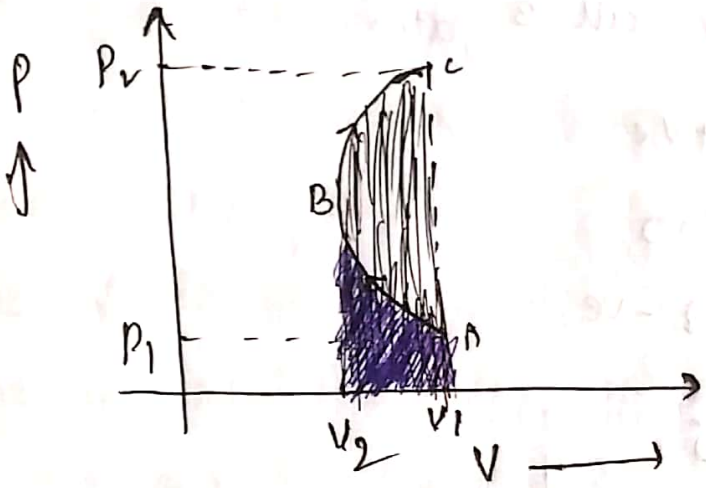
$= \pi \times$ Vol radius $\times$ pressure radius

$$= \pi \times \left(\frac{V_2 - V_1}{2}\right)\left(\frac{P_2 - P_1}{2}\right)$$

$$\boxed{W = \frac{\pi}{4}(V_2 - V_1)(P_2 - P_1) \text{ atm Lit}}$$

② what is the workdone by the gas in the graph in the path $A \longrightarrow B \longrightarrow C$

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$W_{A \to B} = -ve$

$W_{B \to C} = +ve$

∴ net work done by the gas,

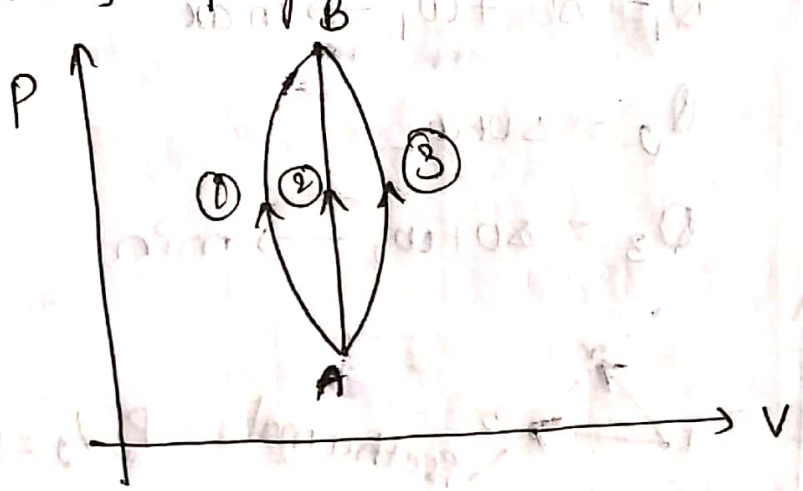
$$W = W_{A \to B} + W_{B \to C}$$

⇒ W = area of semi circle

$$= \frac{\pi}{2} \times (V_2 - V_1) \times \left(\frac{P_2 - P_1}{2}\right)$$

$$\boxed{W = \frac{\pi}{4}(V_2 - V_1)(P_2 - P_1)}$$

(3) Identify the path in which molecular heat capacity of gas is max.



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ΔU = same for all 3 paths

$W_1 \rightarrow +ve$

$W_2 \rightarrow 0$

$W_3 \rightarrow -ve$

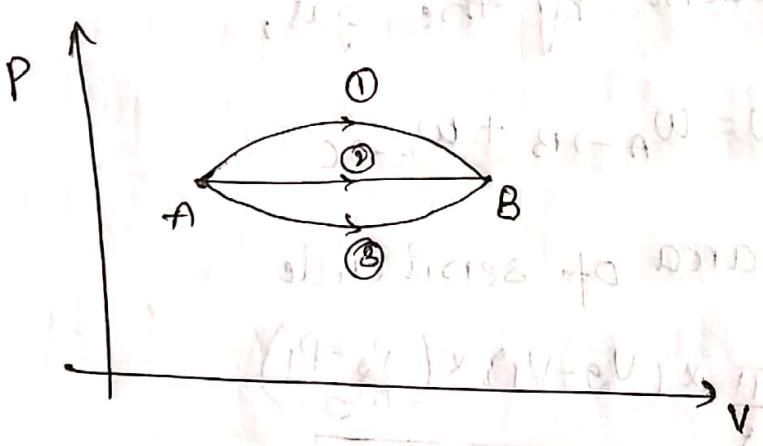
$\Delta U = Q - W$

$Q_1 = \Delta U + W_1 \rightarrow max$

$Q_2 = \Delta U$

$Q_3 = \Delta U - W_3 \rightarrow min$

(4)



Here ΔU = same for all particles

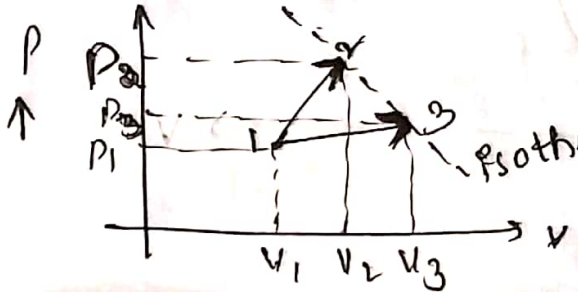
& $W_1 > W_2 > W_3$

now, $Q_1 = \Delta U + W_1 \rightarrow max$

$Q_2 = \Delta U + W_2$

$Q_3 = \Delta U + W_2 \rightarrow min$

(5)



$P_2V_2 = P_3V_3$

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A gas takes path in 2 thermal processes in which it is heated from same initial state to the same final temp. The processes are shown on the PV diagram by st. lines $1 \to 2$ & $1 \to 3$ indicate the process in which the amount of heat supplied to the gas is larger.

Sol: ΔU = same along both the path $1 \to 3$ & $1 \to 2$

now, $W_{1\to 2}$ = area of trapezium

$$= \frac{1}{2}(P_1 + P_2)(V_2 - V_1)$$

$$= \frac{1}{2}(P_1V_2 + P_2V_2 - P_1V_1 - P_2V_1)$$

& $W_{1\to 3} = \frac{1}{2}(P_1 + P_3)(V_3 - V_1)$

$$= \frac{1}{2}(P_1V_3 - P_1V_1 + P_3V_3 - P_3V_1)$$

now, $W_{1\to 2} - W_{1\to 3}$

$$= \frac{1}{2}(P_1V_2 + P_2V_2 - P_1V_1 - P_2V_1 - P_1V_3 + P_1V_1 - P_3V_3 + P_3V_1)$$

$$= \frac{1}{2}(P_1(V_2 - V_3) + V_1(P_3 - P_2))$$

$W_{1\to 2} - W_{1\to 3} < 0$

$$\boxed{W_{1\to 2} < W_{1\to 3}}$$

& from 1st law of thermodynamics,

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$$Q_{1\to 2} = \Delta U + W_{1\to 2}$$

$$Q_{1\to 3} = \Delta U + W_{1\to 3}$$

$$Q_{1\to 2} - Q_{1\to 3} = W_{1\to 2} - W_{1\to 3}$$

$$< 0$$

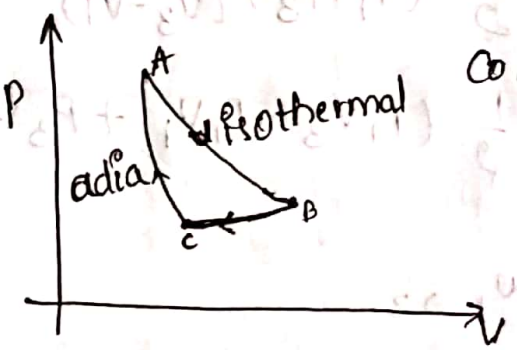
$$\boxed{Q_{1\to 2} < Q_{1\to 3}}$$

∴ Amount of heat supplied is more in path $1 \to 3$.

Conversion of Graphs:

① The sence of cyclic process in P–T diagram is always opposite to the sence of cyclic process in V–T diagram.

Eg:



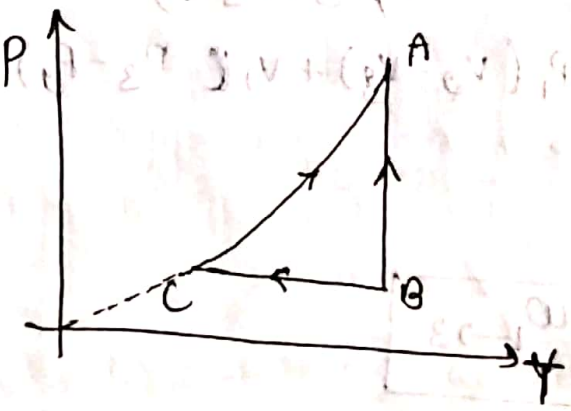
Convert it into

① P–T diagram

② V–T diagram

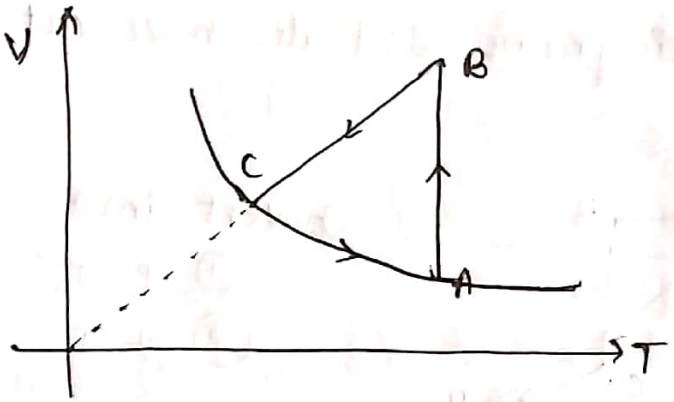
Sol:

P–T diagram:

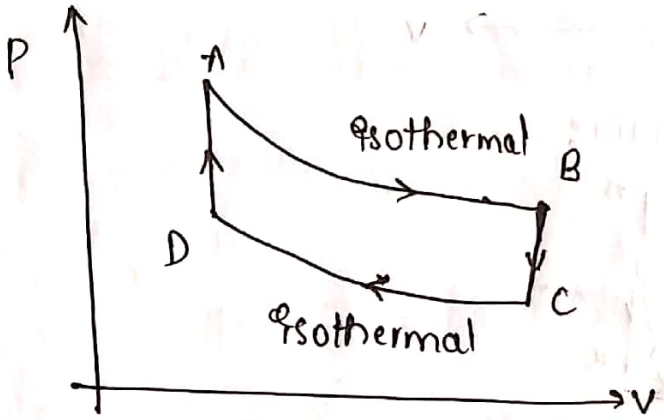


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V-T diagram :



Eg:



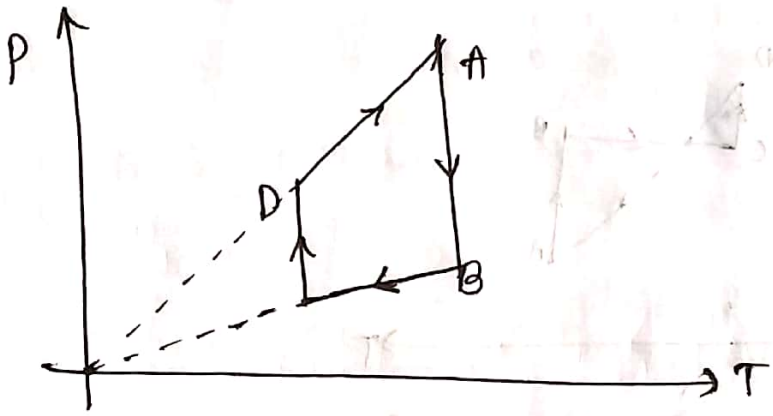
Convert into

(1) P-T diagram

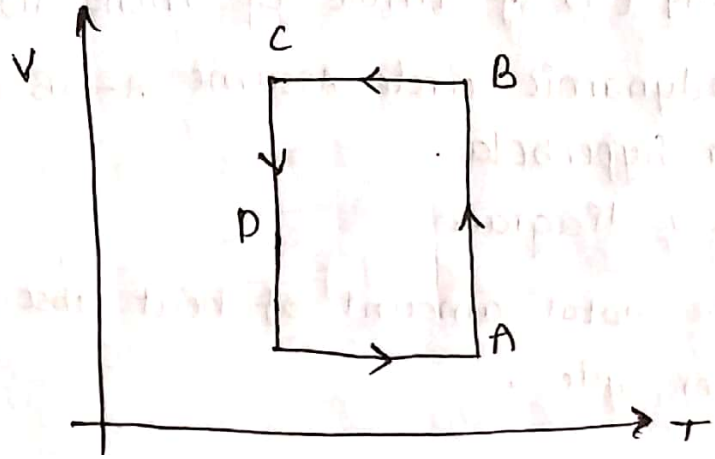
(2) V-T diagram

Sol:

P-T diagram :



V-T diagram :

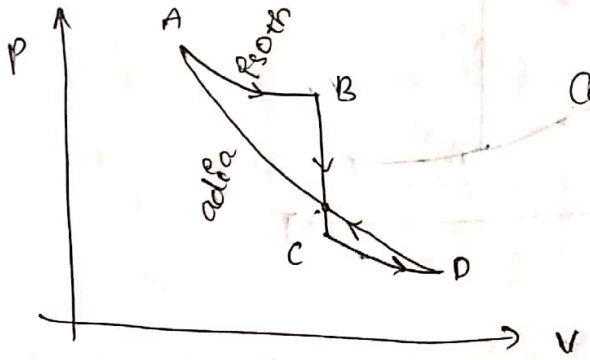


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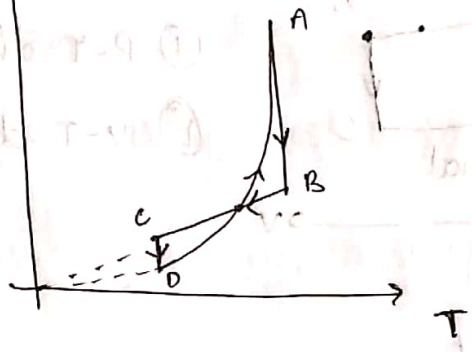
* on a P-V diagram, If we have a cut pt then P-T & V-T diagrams will also have cut pt.

Eg!

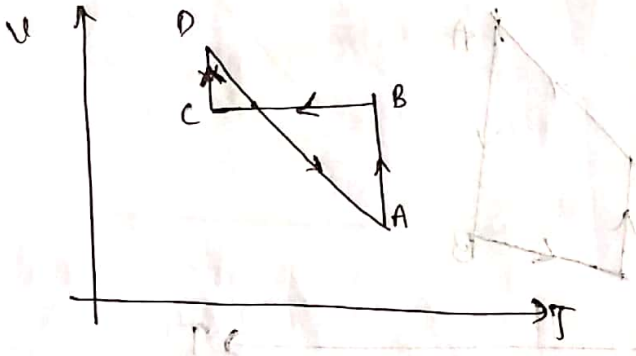


Convert into
① P-T
② V-T.

Sol: P-T diagram:



V-T diagram:



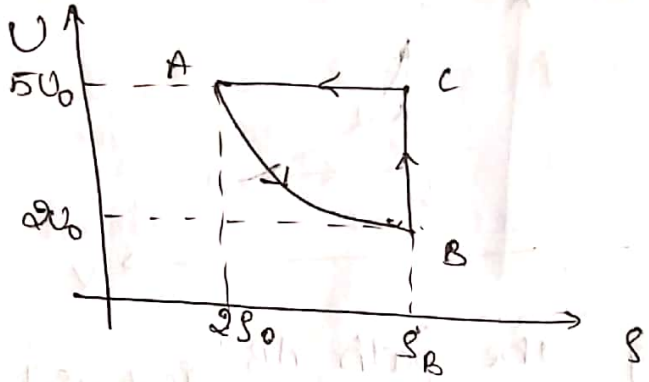
Eg: Figure shows the variation of internal energy (U) with density (ρ) of 1 mole of mono-atomic gas for the thermodynamic circle. Assume A → B to be part of rectangular hyperbola.

① Draw P-V diagram

② Find the total amount of heat absorbed by the system per cycle.

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(3) work done in process $A \rightarrow B$.



Sol:

for path $A \rightarrow B$:

$U\rho = \text{const}$ (rectangular hyperbola)

$$\frac{f}{2} nRT\rho = \text{const}$$

$$\frac{f}{2} nRT \times \frac{m}{V} = \text{const}$$

$$\frac{f}{2} mp = \text{const}$$

$$\boxed{p = \text{const}}$$

$\therefore A \rightarrow B$ is isobaric process.

for path $B \rightarrow C$:

$$\rho = \text{const}$$

$$\frac{m}{V} = \text{const} \Rightarrow \boxed{V = \text{const}}$$

$\therefore B \rightarrow C$ is isochoric process.

for CA :

$$U = \text{const}$$

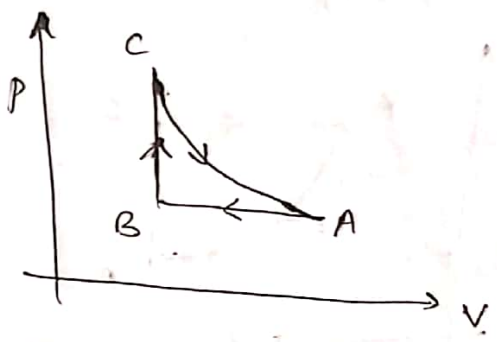
$$\frac{f}{2} nRT = \text{const}$$

$$\boxed{T = \text{const}}$$

$\Rightarrow$ path is isothermal process.

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now,



Along the path AB : [Isobaric]

(a) $\Delta U_{AB} = nC_V (T_B - T_A)$

$= \cancel{1} \times \frac{3R}{\cancel{2}} \left(\frac{\overset{2}{\cancel{4}}U_0}{3\cancel{R}} - \frac{\overset{5}{\cancel{10}}U_0}{3\cancel{R}} \right)$

$\boxed{\Delta U_{AB} = -3U_0}$

(Or)

$\Delta U_{AB} = U_B - U_A$

$= 2U_0 - 5U_0$

$\Delta U_{AB} = -3U_0$

(b) $Q_{AB} = nC_P (T_B - T_A)$

$= \cancel{1} \left(\frac{5\cancel{R}}{\cancel{2}} \right) \left(\frac{\overset{2}{\cancel{4}}U_0}{3\cancel{R}} - \frac{\overset{5}{\cancel{10}}U_0}{3\cancel{R}} \right)$

$= 5 (-U_0)$

$\boxed{Q_{AB} = -5U_0}$

(c) $W_{AB} = Q_{AB} - \Delta U_{AB}$

$= -5U_0 + 3U_0$

$\boxed{W_{AB} = -2U_0}$

$U_B = \frac{f}{2} nRT_B$

$2U_0 = \frac{3}{2} nRT_B$

$\boxed{T_B = \frac{4U_0}{3R}}$

$U_A = \frac{f}{2} nRT_A$

$5U_0 = \frac{3}{2} nRT_A$

$\boxed{T_A = \frac{10U_0}{3R}}$

$\boxed{T_A = T_C}$

Along the path BC: [Isochoric]

(a) $W_{BC} = 0$

(b) $Q_{BC} = \Delta U_{BC} = U_C - U_B = 3U_0$

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(or) $Q_{BC} = nC_V(T_C - T_B)$

$= 1 \times \frac{3R}{2}\left(\frac{100_0}{3R} - \frac{4U_0}{3R}\right)$

$$\boxed{Q_{BC} = 3U_0}$$

Along the path CA :

[Isothermal]

$\Delta U_{CA} = 0$

$Q_{CA} = W_{CA} = nRT_C \ln(V_A/V_C)$

$Q_{CA} = R\left(\frac{10U_0}{3R}\right) \ln\left(\frac{8_C}{8_A}\right)$

$= \frac{10U_0}{3} \ln\left(\frac{5\%_0}{2\%}\right)$

$$\boxed{Q_{CA} = \frac{10U_0}{3} \ln(5/2)}$$

* For a complete cycle,

$$\boxed{\Delta U = 0}$$

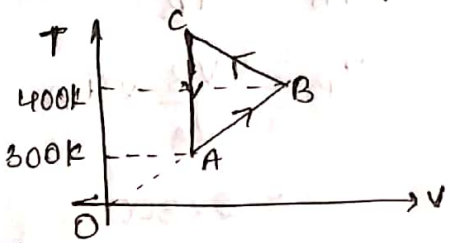
& heat absorbed by gas

$Q = Q_{AB} + Q_{BC} + Q_{CA}$

$= -5U_0 + 3U_0 + \frac{10U_0}{3} \ln(5/2)$

$$\boxed{Q = \frac{10U_0}{3} \ln(5/2) - 2U_0}$$

Eg: Figure shows a cyclic process containing 1 mole of ideal gas, a total of 1000 J of heat is withdrawn from a gas in complete cycle. Calculate workdone by the gas during the process BC.



Sol: Given $n = 1$ & $Q = -1000$ J

for a cyclic process, $\boxed{\Delta U = 0}$

$Q = \Delta U^{0} + W$

$$\boxed{Q = W}$$

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$-1000 = W_{AB} + W_{BC} + W_{CA}$ —(1)

now process AB → isobaric

$W_{AB} = nR(T_B - T_A) = (1)R(400-300) = 831.4\,J$

process AC → isochoric

$\boxed{W_{AC} = 0}$

(1) ⇒ $-1000 = 831.4 + W_{BC}$

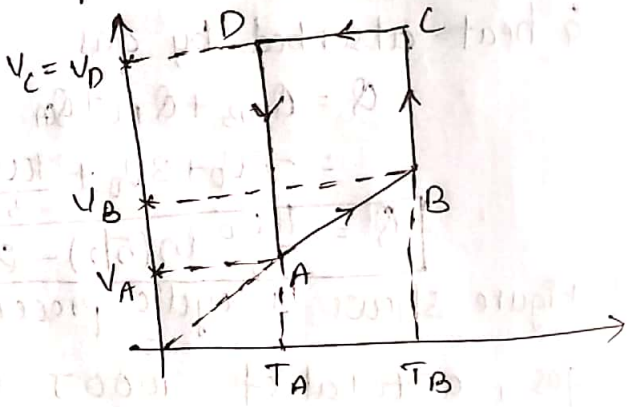
$\boxed{W_{BC} = -1831.4\,J}$

Eg: 2 moles of ideal mono-atomic gas is taken through a cyclic process starting from A as shown. Volume ratios are $V_B : V_A = 2$, $V_D : V_A = 4$. If $T_A = 27°C$. Calculate

(1) T_B

(2) Heat absorbed (or) released by gas in each process.

(3) Total work done during the complete cycle.



Sol: Given $n = 2$, $C_V = \frac{3R}{2}$,

$C_P = \frac{5R}{2}$

(1) Along path AB:

pressure = const

$\frac{V_A}{T_A} = \frac{V_B}{T_B}$

$T_B = \frac{V_B}{V_A} \times T_A$

$= 2 \times 300$

$= 600K$

$\boxed{T_B = 327°C}$

$T_A = 27 + 273 = 300K$

(2) Along path AB:

$Q_{AB} = nC_P(T_B - T_A)$

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$= 2 \times \frac{5R}{2} (300)$

$$\boxed{Q_{AB} = 1500R}$$

& $\Delta U_{AB} = nC_V(T_B - T_A)$

$= 2 \times \frac{3R}{2} (300)$

$$\boxed{\Delta U_{AB} = 900R}$$

$W_{AB} = Q_{AB} - \Delta U_{AB}$

$= 1500R - 900R$

$$\boxed{W_{AB} = 600R}$$

Along the path BC :

$\Delta U_{BC} = 0$

$Q_{BC} = W_{BC} = nRT_B \ln\left(\frac{V_C}{V_B}\right)$

$= (2)(R)(600) \ln(2)$

$$\boxed{Q_{BC} = W_{BC} = 1200R \ln 2}$$

$$\boxed{Q_{BC} = 831.6R}$$

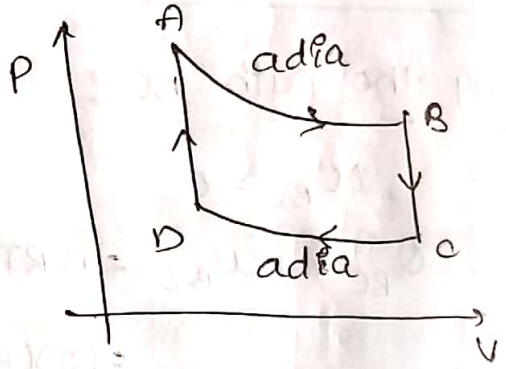
Along the path CD :

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Eg: 1 mole of mono-atomic ideal gas undergoes a cyclic process as shown. Calculate ① work done by the gas in $A \to B$.

② Heat lost in $B \to C$.

③ Temperature T_B.



Given $T_A = 1000K$

$$P_B = \frac{2}{3} P_A$$

$$P_C = \frac{1}{3} P_A$$

take $\left(\frac{2}{3}\right)^{2/5} = 0.85$

Sol: $n = 1$, $\gamma = 5/3$

Along the path AB:-

$$P^{1-\gamma} T^{\gamma} = \text{const}$$

$$\left(\frac{P_A}{P_B}\right)^{1-\gamma} = \left(\frac{T_B}{T_A}\right)^{\gamma}$$

$$\frac{P_A}{P_B} = \left(\frac{T_B}{T_A}\right)^{\gamma/1-\gamma}$$

$$\frac{P_A}{\frac{2}{3}P_A} = \left(\frac{T_B}{1000}\right)^{-5/2}$$

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$$\left(\frac{3}{2}\right)^{-2/5} \times 1000 = T_B$$

$$\left(\frac{2}{3}\right)^{2/5} \times 1000 = T_B$$

$$\boxed{T_B = 850K}$$

now,

$$W_{AB} = \frac{nR}{\gamma - 1}(T_A - T_B)$$

$$= \frac{1 \times R}{\frac{2}{3}}(150)$$

$$\boxed{W_{AB} = 225R}$$

& $\boxed{Q_{AB} = 0}$

$$\Delta U_{AB} = -W_{AB}$$

$$\boxed{\Delta U_{AB} = -225R}$$

$$nC_V \Delta T = -225R$$

$$1 \times \frac{3R}{2} \Delta T = -225R$$

$$\boxed{\Delta T = -150K}$$

Along BC :- [Isochoric]

$$\frac{P_B}{T_B} = \frac{P_C}{T_C}$$

$$\frac{\frac{2}{3}P_A}{\frac{1}{3}P_A} = \frac{T_B}{T_C}$$

$$2 = \frac{850}{T_C} \Rightarrow \boxed{T_C = 425K}$$

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$W_{BC} = 0$

$Q_{BC} = \Delta U_{BC}$

$= nC_V (T_C - T_B)$

$= 1 \times \frac{3R}{2} (-425)$

$= -1.5 \times 425 R$

$\boxed{Q_{BC} = -637.5R}$

Along CD :-

$P_C^{1-\gamma} T_C^{\gamma} = P_D^{1-\gamma} T_D^{\gamma}$

$\frac{P_C}{P_D} T_C^{\gamma/1-\gamma} = T_D^{\gamma/1-\gamma}$

$\frac{2}{3} * = \left(\frac{T_D}{425}\right)^{-5/2}$

$\left(\frac{2}{3}\right)^{-2/5} = \frac{T_D}{425}$

$\frac{425}{0.85} = T_D$

$T_D = \frac{425}{85} \times 100$

$\boxed{T_D = 500K}$

we have;

$P_A V_A^{\gamma} = P_B V_B^{\gamma}$ -①

$P_C V_C^{\gamma} = P_D V_D^{\gamma}$ -②

$\frac{②}{①} \Rightarrow \frac{P_A V_A^{\gamma}}{P_D V_D^{\gamma}} = \frac{P_B^{\gamma} V_B^{\gamma}}{P_C V_C^{\gamma}}$

$\boxed{\frac{P_C}{P_D} = \frac{2}{3}}$

$Q_{CD} = 0$

$W_{CD} = \frac{nR}{\gamma - 1} (T_C - T_D)$

$= \frac{1 \times R}{2/3} (-75)$

$= -\frac{3 \times 75 R}{2} = -1.5 \times 75 R = -112.5R$

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& $\Delta U_{CD} = -W_{CD}$

$$\boxed{\Delta U_{CD} = 112.5R\ J}$$

Along DA :- [isochoric]

$W_{DA} = 0$

$$Q_{DA} = \Delta U_{DA} = nC_V(T_A - T_D)$$

$$= 1 \times \frac{3R}{2}(250)$$

$$\boxed{Q_{DA} = 750R}$$

Finally,

$$Q = W = 0 + (-637.5R) + 0 + 750R$$

$$\boxed{Q = 112.5R\ J}$$

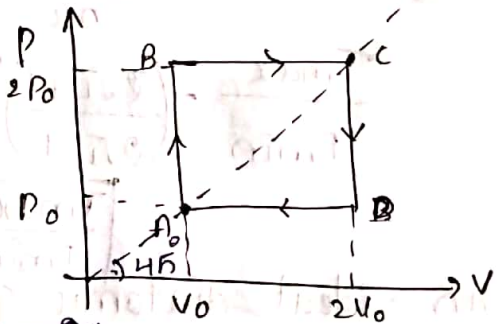
To find Maximum Temperature!

$$T_A = \frac{P_0V_0}{nR} \quad \text{– min}$$

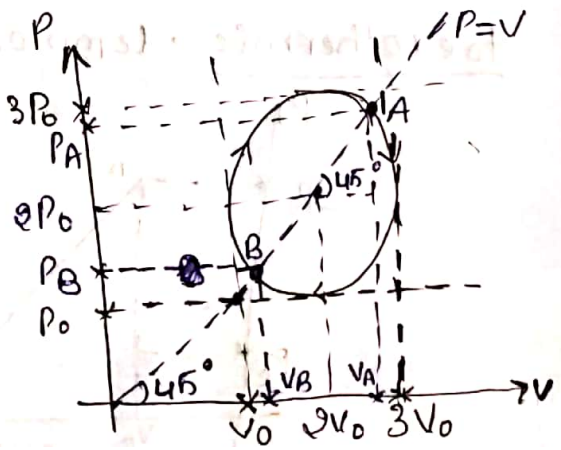
$$T_B = \frac{(2P_0)V_0}{nR} = \frac{2P_0V_0}{nR}$$

$$T_C = \frac{(2P_0)(2V_0)}{nR} = 4\frac{P_0V_0}{nR} \quad \text{– max}$$

$$T_D = \frac{P_0(2V_0)}{nR} = \frac{2P_0V_0}{nR}$$



Eg! Find the ratio of max & min Temp of gas in a cycle.



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co-ordinates of A &

$$V_A = 2V_0 + V_0 \cos 45^\circ$$

$$V_A = \left(\frac{2\sqrt{2}+1}{\sqrt{2}}\right)V_0$$

& $P_A = 2P_0 + P_0 \cos 45^\circ$

$$= \left(\frac{2\sqrt{2}+1}{\sqrt{2}}\right)P_0$$

now, $T_A = T_{max} = \frac{P_A V_A}{nR} = \frac{P_0 V_0}{2nR}(2\sqrt{2}+1)^2$

Co-ordinates of B &

$$V_B = 2V_0 - \frac{V_0}{\sqrt{2}} = \left(\frac{2\sqrt{2}-1}{\sqrt{2}}\right)V_0$$

$$P_B = \left(\frac{2\sqrt{2}-1}{\sqrt{2}}\right)P_0$$

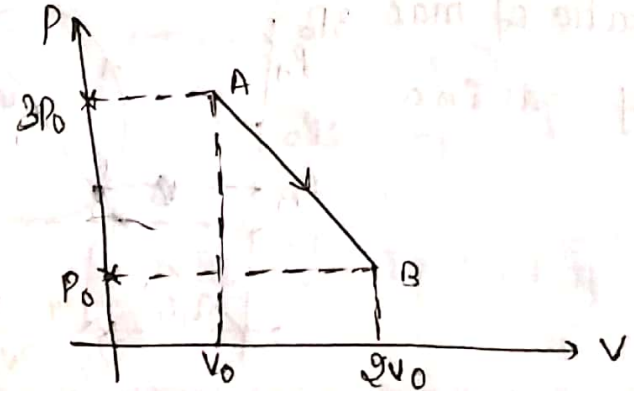
$$T_B = T_{min} = \frac{P_B V_A}{nR} = \frac{P_0 V_0}{2nR}(2\sqrt{2}-1)^2$$

$$\frac{T_{max}}{T_{min}} = \left(\frac{2\sqrt{2}+1}{2\sqrt{2}-1}\right)^2 = \left(\frac{4+\sqrt{2}}{4-\sqrt{2}}\right)^2 = \left(\frac{(4+\sqrt{2})^2}{16-2}\right)^2 = \left(\frac{16+2+8\sqrt{2}}{12}\right)^2$$

$$\boxed{\frac{T_{max}}{T_{min}} = \left(\frac{18+8\sqrt{2}}{12}\right)^2}$$

V.V.V Imp

Eg. An ideal diatomic gas undergoes a process A → B for which P–V diagram is as shown. Temp at A is T_0 when the process changes from endothermic to exothermic. temp of gas is $\frac{175\alpha T_0}{864}$. Find α.



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Sol:

Temp of gas at A, $T_0 = \frac{3P_0V_0}{nR}$

* eqn of straight line,

$P = mV + C$

$3P_0 = mV_0 + C$

& $P_0 = m(2V_0) + C$

$2P_0 = -mV_0$

$\boxed{m = -\frac{2P_0}{V_0}}$ & $3P_0 = -\frac{2P_0}{V_0} \times V_0 + C$

$\boxed{C = 5P_0}$

eqn,

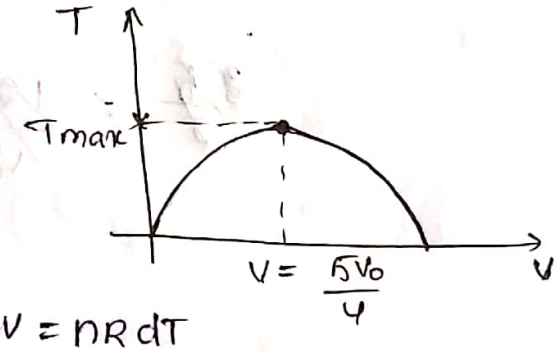
$$\boxed{P = -\frac{2P_0}{V_0}V + 5P_0} \quad -(1)$$

now, $PV = nRT$

$$\left(-\frac{2P_0}{V_0}V + 5P_0\right)V = nRT$$

$$-\frac{2P_0}{V_0}V^2 + 5P_0V = nRT \quad -(2)$$

To find max temp,



$\frac{dT}{dV} = 0$

$$-\frac{2P_0}{V_0}(2V)\cdot dV + 5P_0\,dV = nR\,dT$$

$$-\frac{4P_0V}{V_0} + 5P_0 = nR\left(\frac{dT}{dV}\right)$$

$$\frac{\cancel{4}P_0V}{V_0} = \cancel{5}P_0$$

$$\Rightarrow \boxed{\frac{5V_0}{4} = V} \quad \& \quad P = -\frac{2P_0}{V_0} \times \frac{5V_0}{4 \times 2} + 5P_0 = \frac{5P_0}{2}$$

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$$T_{max} = \frac{PV}{nR} = \frac{\frac{5P_0}{2} \times \frac{5V_0}{4}}{nR} = \frac{25P_0V_0}{8nR}$$

(or) sub $V = \frac{5V_0}{4}$ in eqn ②, we get T_{max}.

initially gas absorbs heat and then releases heat

(endothermic) $dQ > 0$ (exothermic) $dQ < 0$

at critical pt,

$$\boxed{dQ = 0}$$

$$dU + dW = 0$$

$$nC_V dT + P dV = 0$$

$$\frac{5R}{2} \times \left(-\frac{4P_0V}{V_0} + 5P_0\right) \frac{dV}{dR} + \left(-\frac{2P_0V}{V_0} + 5P_0\right) dV = 0$$

$$\frac{5}{2}\left(5P_0 - \frac{4P_0V}{V_0}\right) + 5P_0 - \frac{2P_0V}{V_0} = 0$$

$$\frac{25}{2}P_0 + 5P_0 = \frac{10P_0V}{V_0} + \frac{2P_0V}{V_0}$$

$$\frac{35}{2} = \frac{12V}{V_0}$$

$$\boxed{\frac{35V_0}{24} = V}$$

at this volume,

$$P = -\frac{2P_0}{V_0}\left(\frac{35V_0}{24}\right) + 5P_0$$

$$= -\frac{35P_0}{12} + 60P_0$$

$$\boxed{P = \frac{25P_0}{12}}$$

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∴ Temp at which process changes from endothermic to exothermic,

$$T = \frac{PV}{nR} = \frac{\frac{35V_0}{24} \times \frac{25P_0}{12}}{nR}$$

$$\frac{35 \times 25 \times P_0 V_0}{24 \times 12} \times \frac{1}{nR} = \frac{35 \times 25}{24 \times 12} \frac{T_0}{3}$$

$$\boxed{\alpha = 5}$$

1/11/24 Friday

An ideal gas is made to undergo thermodynamic process given by $V \propto T^{\gamma}$. Find heat capacity of gas for this process.

Sol:

$$V \propto T^{\gamma}$$

$$PV = nRT$$

$$T = \frac{PV}{nR}$$

$$\propto \frac{P^{\gamma} V^{\gamma}}{n^{\gamma} R^{\gamma}}$$

$$P^{\gamma} V = \text{const}$$

$$P V^{1/\gamma} = \text{const}$$

comparing with $PV^{x} = \text{const}$

$$\boxed{x = 1/\gamma}$$

we have, $C = C_V + \frac{R}{1-x}$

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$$C = C_V + \frac{R}{1-1/\gamma}$$

$$\boxed{C = C_V + 2R}$$

$$C = \frac{R}{\gamma - 1} + 2R$$

Eg: 1 mol of an ideal gas whose adiabatic coefficient γ undergoes a process given by $P = P_0 + \frac{\alpha}{V}$. Find

1) Specific heat capacity of gas as a function of volume.

2) change in internal energy & amount of work done by the gas when expands from V_1 to V_2.

Soln

$$P = P_0 + \frac{\alpha}{V} \quad ; \quad P_0, \alpha \text{ — consts}$$

$$dP = -\frac{\alpha}{V^2} \cdot dV$$

we have,

$$PV = nRT$$

$$P\,dV + V\,dP = nR\,dT$$

$$\left(P_0 + \frac{\alpha}{V}\right) dV + V\left(-\frac{\alpha}{V^2} \cdot dV\right) = nR\,dT$$

$$\left(P_0 + \frac{\alpha}{V} - \frac{\alpha}{V}\right) \cdot dV = nR\,dT$$

$$\boxed{\frac{dV}{dT} = \frac{nR}{P_0}}$$

we have,

from 1st law of Thermodynamics,

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$$dQ = dU + dW$$

$$nC\,dT = nC_V\,dT + P\,dV$$

$$C = C_V + \frac{P\,dV}{n\,dT}$$

$$= C_V + \frac{\left(P_0 + \frac{\alpha}{V}\right)}{\cancel{D}V} \times \frac{\cancel{D}R}{P_0}$$

$$= C_V + \left(\frac{P_0 + \frac{\alpha}{V}}{P_0}\right)R$$

$$\boxed{C = C_V + \left(P_0 + \frac{\alpha}{V}\right)\frac{R}{P_0}}$$

Change in internal energy,

$$dU = nC_V\,dT$$

$$= \cancel{n}C_V \frac{P_0}{\cancel{n}R}\,dV$$

$$dU = \frac{P_0 C_V}{R}\,dV$$

$$\Delta U = \int_{V_1}^{V_2} dU$$

$$\Delta U = \frac{P_0 C_V}{R}\int_{V_1}^{V_2} dV$$

$$\boxed{\Delta U = \frac{P_0 C_V}{R}(V_2 - V_1)}$$

Work done,

$$W = \int_{V_1}^{V_2} P\,dV = \int_{V_1}^{V_2}\left(P_0 + \frac{\alpha}{V}\right)dV$$

$$\boxed{W = P_0(V_2 - V_1) + \alpha \ln(V_2/V_1)}$$

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Ex: Molar heat capacity of an ideal gas 1) $C = C_V$ const,

2) $C = C_V + \beta V$ 3) $C = C_V + \frac{\alpha}{p}$ where $\alpha, \beta, a \rightarrow$ consts

Find eqn of the process for an ideal gas in terms of variables, T, P.

Sol:

we have,

$$dQ = dU + dW$$

$$nC\,dT = nC_V\,dT + P\,dV$$

$$\boxed{C = C_V + \frac{P\,dV}{n\,dT}}$$

and $P = \frac{nRT}{V}$

$$C = C_V + \left(\frac{RT}{V}\right)\frac{dV}{dT} \quad -(a)$$

Given, (1) $C = C_V + \alpha T$

$$\alpha T = \frac{RT}{V} \cdot \frac{dV}{dT}$$

$$\int \frac{\alpha}{R} \cdot dT = \int \frac{1}{V} \cdot dV$$

$$\ln V = \frac{\alpha}{R}T + \text{const} \quad \hookrightarrow \ln k$$

$$\ln V - \ln k = \frac{\alpha}{R}T$$

$$\ln(V/k) = \frac{\alpha T}{R}$$

$$\frac{V}{k} = e^{\frac{\alpha T}{R}}$$

$$\boxed{Ve^{-\alpha T/R} = \text{const}}$$

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(2) $C = C_V + \beta V$

$$\beta V = \frac{RT}{V} \cdot \frac{dV}{dT}$$

$$\int \frac{\beta}{RT} \cdot dT = \int \frac{1}{V} \cdot dV$$

$$\frac{\beta}{R} \cdot \ln T = -\frac{1}{V} + \text{const}$$

$$\ln T = -\frac{R}{\beta V} + \text{const} \quad \hookrightarrow \ln k$$

$$\ln(T/k) = -\frac{R}{\beta V}$$

$$\frac{T}{k} = e^{-R/\beta V}$$

$$\boxed{T e^{R/\beta V} = \text{const}}$$

(3) $C = C_V + ap$

$$ap = \frac{p\,dV}{n\,dT}$$

$$an \cdot dT = dV$$

$$\int an\,dT = \int dV$$

$$\boxed{V = anT + \text{const}}$$

Ex: An ideal gas has a molar heat capacity C_V at const volume. Find molar heat capacity of this gas as a function of its volume V, if the gas undergoes the following proces ① $T = T_0 e^{\alpha V}$

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(9) $P = P_0 e^{\alpha V}$.

Sol:

we have,

$$C = C_V + \frac{P dV}{n dT} \quad \& \quad P = \frac{nRT}{V}$$

(1) Given, $T = T_0 e^{\alpha V}$

$$dT = T_0 \alpha e^{\alpha V} . dV$$

$$\frac{dV}{dT} = \frac{1}{T_0 \alpha e^{\alpha V}}$$

$$C = C_V + \frac{P}{n} \times \frac{1}{T_0 \alpha e^{\alpha V}}$$

$$= C_V + \frac{nRT}{Vn} \times \frac{1}{T_0 \alpha e^{\alpha V}}$$

$$\boxed{C = C_V + \frac{R}{\alpha V}}$$

(2) Given, $P = P_0 e^{\alpha V}$

$$\frac{nRT}{V} = P_0 e^{\alpha V}$$

$$nR\,dT = P_0 e^{\alpha V} dV + V P_0 \alpha e^{\alpha V} . dV$$

$$nR\,dT = P_0 e^{\alpha V} dV (1 + \alpha V)$$

$$\frac{nR}{P_0 e^{\alpha V}(1+\alpha V)} = \frac{dV}{dT}$$

$$C = C_V + \frac{P dV}{n dT} = C_V + \frac{P \times nR}{n (P_0 e^{\alpha V})(1+\alpha V)}$$

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$$C = C_v + \frac{R}{1+\alpha v}$$

Ex: Consider 1 mole of an ideal gas whose Vol changes with temp as $v = \frac{\alpha}{T}$ where α = const Find amount of heat required to raise the temp by ΔT if adiabatic coeff is γ.

Given, $v = \frac{\alpha}{T}$

$PV = nRT$

$PV = nR\frac{\alpha}{V}$

$P = \frac{nR\alpha}{V^2}$

now workdone, $W = \int P\,dV$

$$= \int_{V_1}^{V_2} n \cdot \frac{R\alpha}{V^2}\,dV$$

$$= -nR\alpha \left[\frac{1}{V}\right]_{V_1}^{V_2}$$

$$= -nR\alpha \left[\frac{1}{V_2} - \frac{1}{V_1}\right]$$

$$W = -nR\left[\frac{\alpha}{V_2} - \frac{\alpha}{V_1}\right]$$

$$W = -nR(T_2 - T_1)$$

change in internal energy,

$$\Delta U = nC_v(T_2 - T_1)$$

$$\Delta U = \frac{nR}{\gamma - 1}(T_2 - T_1)$$

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from First law of Thermodynamics,

$$Q = \Delta U + W$$

$$Q = \frac{nR}{\gamma - 1}(T_2 - T_1) - nR(T_2 - T_1)$$

$$Q = nR(T_2 - T_1)\left(\frac{1}{\gamma - 1} - 1\right)$$

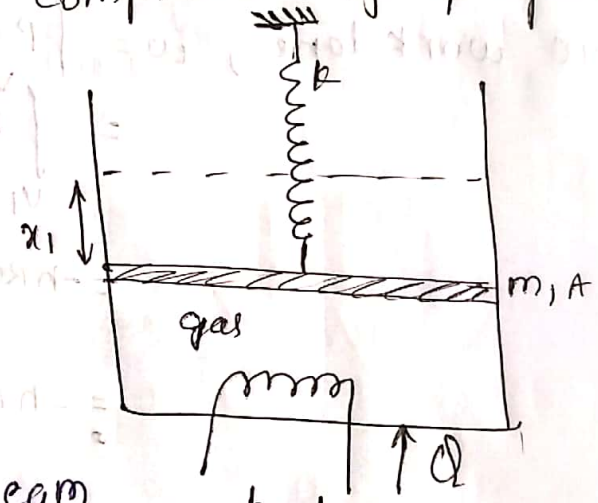
$$\boxed{Q = nR\left(\frac{2 - \gamma}{\gamma - 1}\right)(T_2 - T_1)}$$

Illustration:

Ex: n-moles of mono-atomic gas is taken inside an insulated conductor having frictionless insulated piston. The initial volume of gas = V & spring is initially in relaxed state now by the help of the heater the gas is slowly heated & finally compression of spring = x_1

Sol:

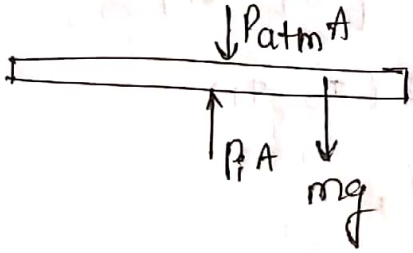
Initial condn:



$$P_i V = nRT_i$$

$$T_i = \frac{P_i V}{nR}$$

since piston is in eqm,



$$P_iA = P_{atm}A + mg$$

$$\boxed{P_i = P_{atm} + \frac{mg}{A}}$$

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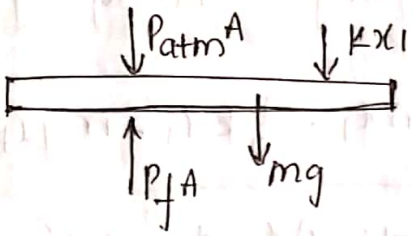
Final conditions

$$P_f V_f = nRT_f$$

$$\boxed{T_f = \frac{P_f V_f}{nR}}$$

here, $V_f = V + Ax_1$

In eqm,



$$P_f A = P_{atm} A + kx_1 + mg$$

$$\boxed{P_f = P_{atm} + \frac{kx_1}{A} + \frac{mg}{A}}$$

now,

change in internal energy of gas,

$$\boxed{\Delta U = nC_v (T_f - T_i)}$$

work done by gas,

① force method :

at any intermediate state,

$$P_{gas} = P_{atm} + \frac{kx}{A} + \frac{mg}{A}$$

work done by the gas,

$$W_{gas} = \int P_{gas} \cdot dV = \int_{x=0}^{x=x_1} P_{gas} A \cdot dx$$

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$$W_{gas} = \int_{x=0}^{x=x_1} (P_{atm}A + kx + mg)\,dx$$

$$\boxed{W_{gas} = P_{atm}x_1 + \frac{kx_1^2}{2} + mgx_1}$$

② work-energy theorem

on a piston,

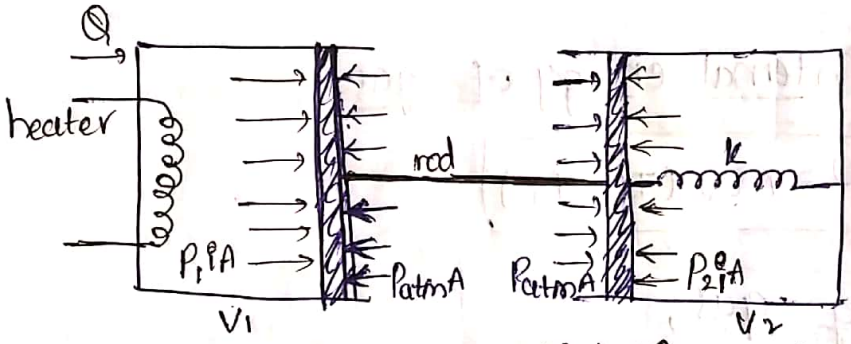
$$\cancel{\Delta K}^{0} = W_{atm} + W_{spring} + W_{gravity} + W_{gas}$$

$$W_{gas} - P_{atm}Ax_1 - \frac{1}{2}kx_1^2 - mgx_1 = 0$$

$$\boxed{W_{gas} = P_{atm}Ax_1 + \frac{1}{2}kx_1^2 + mgx_1}$$

Illustration:

n-moles of diatomic gas enclosed in cylinders ① & ②. Initially spring is in relaxed state.



All are adiabatic surfaces

Initial condn:

For ①,

$$P_{1i}V_1 = nRT_{1i}$$

$$T_{1i} = \frac{P_{1i}V_1}{nR} \quad -①$$

& $$P_{2i}V_2 = nRT_{2i}$$

$$T_{2i} = \frac{P_{2i}V_2}{nR} \quad -②$$

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By force balance,

$$P_{1i} A = P_{atm} A + F$$

& $$P_{atm} A + F = P_{2i} A$$

$$\boxed{P_{1i} = P_{2i}}$$

Final condn:

x = compression of spring

$$V_{2f} = V_2 - Ax$$

$$P_{2f} \times V_{2f} = nRT_{2f}$$

$$\boxed{T_{2f} = \frac{P_{2f} V_{2f}}{nR}}$$

& $$V_{1f} = V_1 + Ax$$

$$P_{1f} V_{1f} = nRT_{1f}$$

$$\boxed{T_{1f} = \frac{P_{1f} V_{1f}}{nR}}$$

Again,

By force balance,

$$P_{1f} A = P_{atm} A + F_1$$

$$P_{atm} A + F_1 = P_{2f} A + kx$$

$$\boxed{P_{1f} A = P_{2f} A + kx}$$

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now,

change in internal energy of system,

$$\Delta U = \Delta U_1 + \Delta U_2$$

$$\Delta U = nC_v(T_{1f} - T_{1i}) + nC_v(T_{2f} - T_{2i})$$

* from work energy theorem,

$$W_{gas} + W_{atm} + W_{spring} = \Delta K \overset{0}{}$$

$$W_{atm} = -P_{atm} A x + P_{atm} A x = 0$$

$$W_{gas} - \frac{1}{2} k x^2 = 0$$

$$\boxed{W_{gas} = \frac{1}{2} k x^2}$$

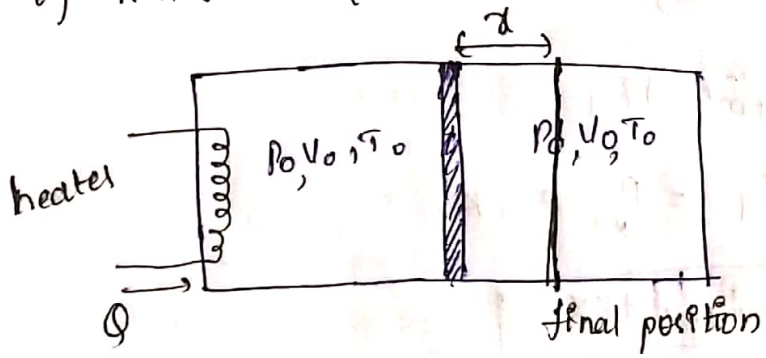
Finally from 1st law of thermodynamics,

$$Q = \Delta U_{system} + W_{gas}$$

$$\boxed{Q = \Delta U_1 + \Delta U_2 + W_{gas}}$$

Illustration:

Insulated cylinder is divided into 2 equal parts by insulating piston. equal moles of mono-atomic gas has be enclosed inside the L.H.S & R.H.S chamber. If Q joules of heat is slowly supplied by the heater to L.H.S till final pressure of R.H.S is P_2.



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Since gas in R·H·S is enclosed by adiabatic walls,

$$P_0 V_0^{\gamma} = P_A V_R^{\gamma}$$

$$V_R^{\gamma} = \frac{P_0}{P_A} V_0^{\gamma}$$

$$\boxed{V_R = V_0 \left(\frac{P_0}{P_A}\right)^{1/\gamma}} \quad \& \quad \boxed{V_R = V_0 - Ax}$$

Again,

$$P_0^{1-\gamma} T_0^{\gamma} = P_R^{1-\gamma} T_R^{\gamma}$$

$$T_R = T_0 \left(\frac{P_0}{P_R}\right)^{\frac{1-\gamma}{\gamma}}$$

For L·H·S :

$$V_L = 2V_0 - V_R = V_0 + Ax$$

$$\frac{P_0 V_0}{T_0} = \frac{P_L V_L}{T_L}$$

$$\boxed{T_L = \frac{P_L V_L T_0}{P_0 V_0}}$$

Again, force balance,

$$P_L A = P_R A$$

$$\boxed{P_L = P_R}$$

now,

① change in internal energy of system,

$$\Delta U = \Delta U_L + \Delta U_R$$

$$= nC_V (T_L - T_0) + nC_V (T_R - T_0)$$

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(2) workdone by the gas on the surrounding,

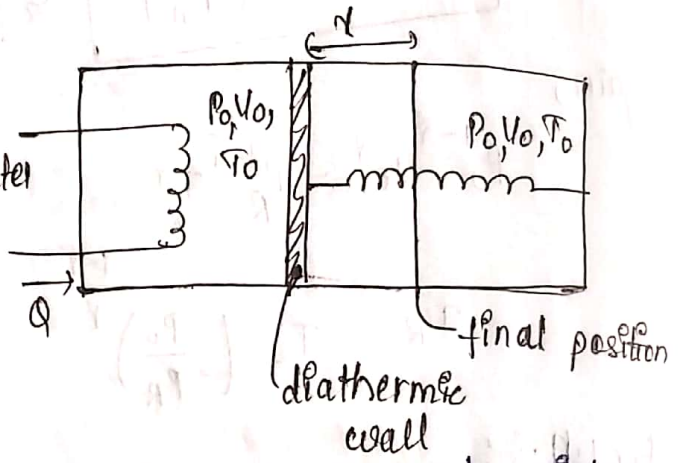
$$\boxed{W_{gas} = 0}$$

(3) $Q = \Delta U + W_{gas}$

$Q = \Delta U_L + \Delta U_R$

Illustration:

Initially spring is released



Final cond^n:

$V_R = V_0 - Ax$

$P_0 V_R = nRT_R$

* Heat flows through piston so temp of both the parts increases and finally becomes T (in thermal eqm)

i.e, $\boxed{T_R = T_L - T}$

* In Mechanical eqm,

$P_L A = P_R A + Kx$

now, $\Delta U = \Delta U_1 + \Delta U_2$

$= nC_V (T_L - T_0) + nCV(T_R - T_0)$

* $W_{gas} + W_{spring} = \Delta K^{0}$

$$\boxed{W_{gas} = \frac{K}{2} x^2}$$

* $(P_L - P_R)A = Kx$

$\int (P_L - P_R) A\, dx = \int Kx\, dx$

$P_L - P_R\, Ax = \frac{K}{2} x^2$

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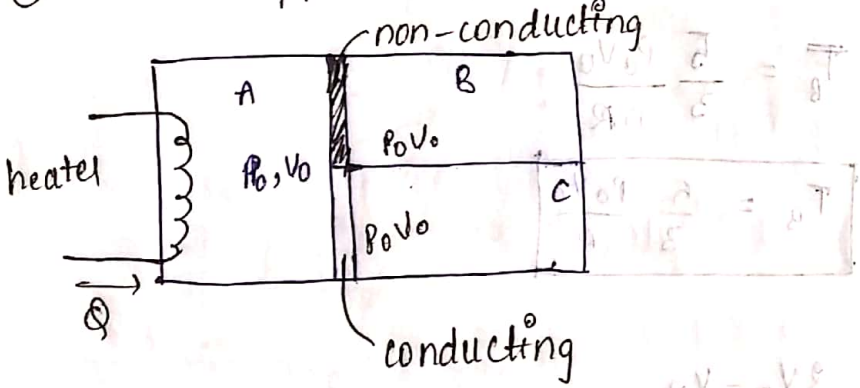
$$Q = \Delta U_L + \Delta U_R + W_{gas}$$

$$= nC_V(T_R - T_0) + nC_V(T_L - T_0) + \frac{k}{2}x^2$$

Ex: Adiabatic container of vol $4V_0$ is divided into 2 equal parts by a rigid fixed wall whose lower half is conducting & upper half is non-conducting. Initially the right portion is divided into 2 equal halves by a freely moving mass less & non-conducting piston. section-A has 2 moles, B & C has each 1 mole of ideal gas with adiabatic exponent $\gamma = 1.5$. If heater in left part is switched on till the final pressure inc becomes $\frac{125}{27}P_0$. Find

(1) final temp in all sections.

(2) heat supplied by the heater.



Sol:

* Since heat flows through conducting wall

$\therefore T_A = T_C$ at all times.

* Since piston is mass less,

$P_B A = P_C A$

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$\boxed{P_B = P_C}$ at all times.

* Gas is enclosed by adiabatic walls from all sides

$\therefore PV^{\gamma} = \text{const}$ given $P_B = P_C = \frac{125P_0}{27}$

$P_0V_0^{\gamma} = P_BV_B^{\gamma}$

$P_0V_0^{\gamma} = \frac{125P_0}{27} V_B^{\gamma}$

$V_B^{\gamma} = \frac{27}{125} V_0^{3/2}$

$V_B = \left(\frac{27}{125}\right)^{2/3} V_0$

$\boxed{V_B = \frac{9}{25} V_0}$

finally, $T_B = \frac{P_BV_B}{nR} = \frac{\frac{125}{27}P_0 \times \frac{9}{25}V_0}{nR}$

$T_B = \frac{5}{3}\frac{P_0V_0}{nR}$

$\boxed{T_B = \frac{5}{3}\frac{P_0V_0}{R}}$

$V_C = 2V_0 - V_B$

$= 2V_0 - \frac{9}{25}V_0$

$\boxed{V_C = \frac{41}{25}V_0}$

$T_C = T_A = \frac{P_CV_C}{nR} = \frac{\frac{125}{27}P_0 \times \frac{41}{25}V_0}{nR}$

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$$T_C = T_A = \frac{205}{27}\frac{P_0V_0}{R}$$

* work done by the system = 0

∴ heat supplied,

$$Q = \Delta U + \cancel{W} \to 0$$

$$Q = \Delta U_A + \Delta U_B + \Delta U_C$$

$$= nC_V(T_A - T_{Ai}) + nC_V(T_B - T_{Bi}) + nC_V(T_C - T_{Ci})$$

$$C_V = \frac{R}{\gamma - 1} = \frac{R}{1/2} = 2R$$

$$T_{Ai} = \frac{P_0(2V_0)}{2(R)}, \quad T_{Bi} = \frac{P_0V_0}{(1)R}, \quad T_{Ci} = \frac{P_0V_0}{(1)R}$$

$$Q = 2(2R)\left(\frac{205}{27}\frac{P_0V_0}{R} - \frac{P_0V_0}{R}\right) + 2R\left(\frac{5}{3}\frac{P_0V_0}{R} - \frac{P_0V_0}{R}\right)$$

$$+ 2R\left(\frac{205}{27}\cdot\frac{P_0V_0}{R} - \frac{P_0V_0}{R}\right)$$

$$= 8 \times 2R\left(\frac{205-27}{27}\right)\frac{P_0V_0}{R} + 2R\left(\frac{2}{3}\right)\frac{P_0V_0}{R}$$

$$= \frac{2}{9}\left(\frac{178}{1}\right)P_0V_0 + \frac{4}{3}P_0V_0$$

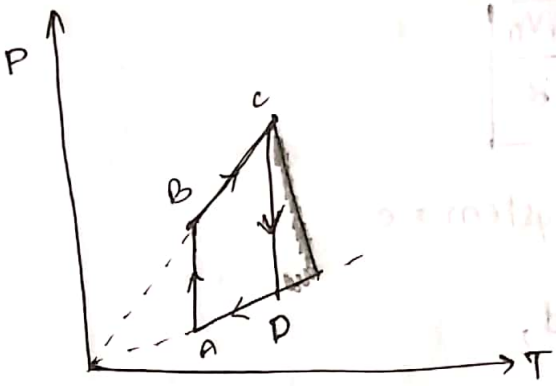
$$= \frac{356}{9}P_0V_0 + \frac{4}{3}P_0V_0$$

$$= \frac{(356+12)}{9}P_0V_0$$

$$\boxed{Q = \frac{368}{9}P_0V_0}$$

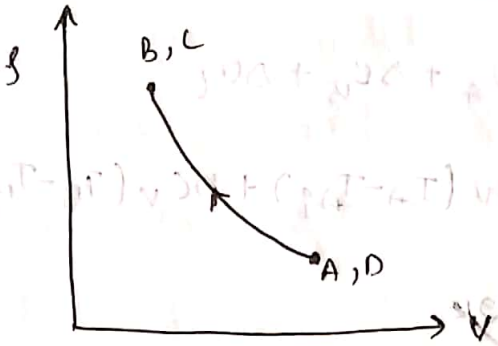
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Ex!



convert into P-V diagram.

Sol!



3|11|24| Monday

Efficiency of a cycle:

For a complete cycle,

$$\Delta U = 0$$

$$Q = \cancel{\Delta U}^{0} + W$$

$$\boxed{Q_{net} = W_{net}}$$

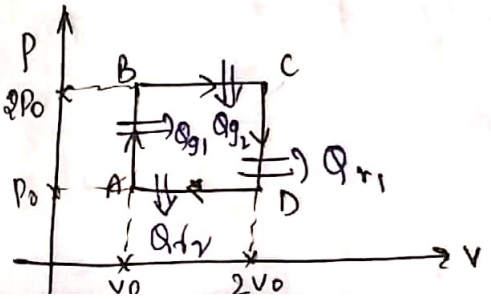
Order of efficiency:

Carnot > diesel > petrol
Engine Engine Engine

* efficiency of a cycle,

$$\eta = \frac{\text{work done by the gas in each cycle}}{\text{Heat absorbed by the gas in that cycle.}}$$

Ex!



Find η, when gas is monoatomic.

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Sol: AB → Isochoric expansion

BC → Isobaric expansion

CD → Isochoric compression

DA → Isobaric compression

$$T_A = \frac{P_0V_0}{nR}, \quad T_B = \frac{2P_0V_0}{nR}, \quad T_C = \frac{4P_0V_0}{nR}, \quad T_D = \frac{2P_0V_0}{nR}$$

Along AB:

$\Rightarrow V = \text{const}$

$$\boxed{W_{AB} = 0}$$

As $P \uparrow \Rightarrow T \uparrow \quad [P \propto T]$

$$Q_{g_1} = nC_V (T_B - T_A)$$

$$= n \times \frac{3R}{2} \left(\frac{P_0V_0}{nR} \right)$$

$$\boxed{Q_{g_1} = \frac{3}{2} P_0V_0}$$

Along BC:

$P = \text{const} \Rightarrow [V \propto T]$

$$W = P(V_C - V_B)$$

$$= (2P_0)(2V_0 - V_0)$$

$$\boxed{W_{BC} = 2P_0V_0}$$

$$Q_{g_2} = nC_p(T_C - T_B)$$

$$= n\left(\frac{5R}{2}\right)\left(\frac{4P_0V_0}{nR} - \frac{2P_0V_0}{nR}\right)$$

$$\boxed{Q_{g_2} = 5P_0V_0}$$

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Along CD:

$V = \text{const}$

$\Rightarrow \boxed{W_{CD} = 0}$

As $P \downarrow \Rightarrow T \downarrow$

$\therefore$ heat of gas is rejected.

Along DA:

$P = \text{const} \quad [V \propto T]$

work done by gas,

$$W_{DA} = P_0 (V_A - V_D)$$

$$= P_0 (V_0 - 2V_0)$$

$$\boxed{W_{DA} = -P_0 V_0}$$

here also gas rejects heat.

$\therefore$ Net work done $= 0 + 2P_0V_0 + 0 - P_0V_0$

$= P_0V_0$

$$\text{efficiency} = \frac{W}{Q_{g_1} + Q_{g_2}} = \frac{P_0V_0}{\left(\frac{3}{2} + 5\right)P_0V_0} = \frac{2}{13}$$

$$\eta = \frac{2}{13} = \frac{200}{13}\% \quad (\eta < 1).$$

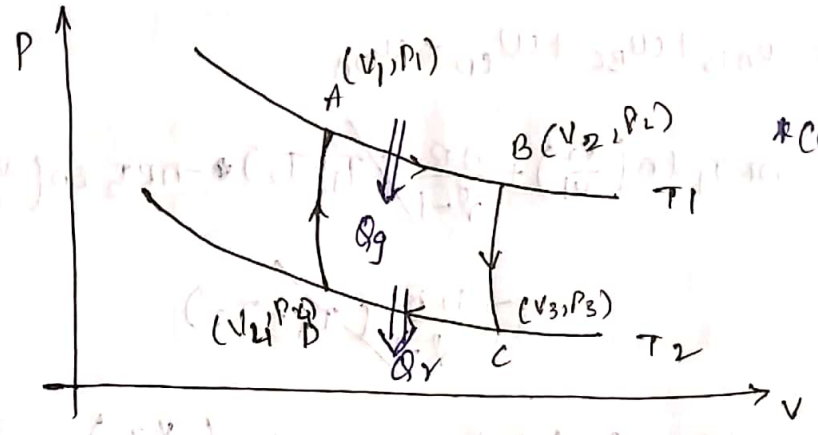
Shortcut:

W_{net} = Area of P-V graph

Q_{g_1}, Q_{g_2} $\Rightarrow$ (we can find) η.

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Efficiency of Carnot Cycle:



carnot engine

AB → Isothermal expansion

BC → adiabatic expansion

CD → Isothermal compression

DA → Adiabatic compression.

Along AB:

$T =$ constant.

$dT = 0 \Rightarrow \boxed{dU = 0}$

$$Q = W_{AB} = nRT_1 \ln\left(\frac{V_2}{V_1}\right)$$

Along BC:

$Q = 0$

& $W = \frac{nR}{\gamma - 1}(T_1 - T_2)$

Along CD:

$$Q_r = W_{CD} = nRT_2 \ln\left(\frac{V_4}{V_3}\right)$$

Along DA:

$Q = 0$

$$W_{DA} = \frac{nR}{\gamma - 1}(T_2 - T_1)$$

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net work done by the gas,

$$W = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$= nRT_1 \ln\left(\frac{V_2}{V_1}\right) + \frac{nR}{\gamma-1}(T_1 - T_2) - nRT_2 \ln\left(\frac{V_3}{V_4}\right) - \frac{nR}{\gamma-1}(T_1 - T_2)$$

$$= nRT_1 \ln\left(\frac{V_2}{V_1}\right) - nRT_2 \ln\left(\frac{V_3}{V_4}\right)$$

efficiency, $\eta = \frac{W}{Q_g} = \frac{nRT_1 \ln\left(\frac{V_2}{V_1}\right) - nRT_2 \ln\left(\frac{V_3}{V_4}\right)}{nRT_1 \ln\left(\frac{V_2}{V_1}\right)}$

$$\eta = 1 - \frac{T_2}{T_1} \frac{\ln\left(\frac{V_3}{V_4}\right)}{\ln\left(\frac{V_2}{V_1}\right)} \quad -(1)$$

Since pts B & C lie on the same adiabatic curve,

$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1}$$

$$\frac{T_1}{T_2} = \left(\frac{V_3}{V_2}\right)^{\gamma-1} \quad -(2)$$

Since pts D & A lies on the same adiabatic curve,

$$T_2 V_4^{\gamma-1} = T_1 V_1^{\gamma-1}$$

$$\frac{T_1}{T_2} = \left(\frac{V_4}{V_1}\right)^{\gamma-1} \quad -(3)$$

From eqns (2) & (3)

$$\left(\frac{V_3}{V_2}\right)^{\gamma-1} = \left(\frac{V_4}{V_1}\right)^{\gamma-1}$$

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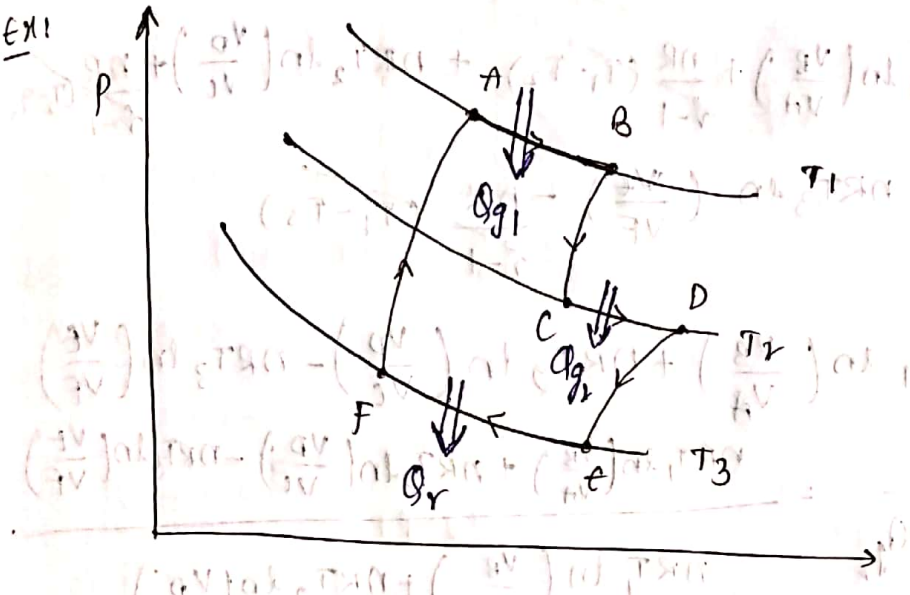
$$\frac{V_3}{V_2} = \frac{V_4}{V_1}$$

∴ eqn (1) becomes,

$$\eta = 1 - \frac{T_2}{T_1} \quad **$$

* $\eta = 1$ when $T_2 = 0K$

Ex1



An ideal gas is taken through a cycle consisting of isothermal & adiabatic process. The isothermal processes are performed at T_1, T_2, T_3. Find η if in each isothermal expansion the gas vol increases in same amount.

Soln

Given $\frac{V_B}{V_A} = k$, $\frac{V_D}{V_C} = k$

$$Q_{g_1} = W_{AB} = nRT_1 \ln\left(\frac{V_B}{V_A}\right)$$

$$W_{BC} = \frac{nR}{\gamma - 1}(T_1 - T_2)$$

$$Q_{g_2} = W_{CD} = nRT_2 \ln\left(\frac{V_D}{V_C}\right)$$

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$$W_{DE} = \frac{nR}{\gamma-1}(T_2 - T_3)$$

$$Q_r = W_{EF} = nRT_3 \ln\left(\frac{V_F}{V_E}\right)$$

$$W_{FA} = \frac{nR}{\gamma-1}(T_3 - T_1)$$

$$W_{net} = W_{AB} + W_{BC} + W_{CD} + W_{DE} + W_{EF} + W_{FA}$$

$$= nRT_1 \ln\left(\frac{V_B}{V_A}\right) + \frac{nR}{\gamma-1}(T_1 - T_2) + nRT_2 \ln\left(\frac{V_D}{V_C}\right) + \frac{nR}{\gamma-1}(T_2 - T_3)$$

$$- nRT_3 \ln\left(\frac{V_E}{V_F}\right) - \frac{nR}{\gamma-1}(T_1 - T_3).$$

$$W_{net} = nRT_1 \ln\left(\frac{V_B}{V_A}\right) + nRT_2 \ln\left(\frac{V_D}{V_C}\right) - nRT_3 \ln\left(\frac{V_E}{V_F}\right)$$

$$\eta = \frac{W}{Q_{g_1} + Q_{g_2}} = \frac{nRT_1 \ln\left(\frac{V_B}{V_A}\right) + nRT_2 \ln\left(\frac{V_D}{V_C}\right) - nRT_3 \ln\left(\frac{V_E}{V_F}\right)}{nRT_1 \ln\left(\frac{V_B}{V_A}\right) + nRT_2 \ln\left(\frac{V_D}{V_C}\right)}$$

Again, $T_1 V_B^{\gamma-1} = T_2 V_C^{\gamma-1}$

$$T_2 V_D^{\gamma-1} = T_3 V_E^{\gamma-1}$$

$$T_3 V_F^{\gamma-1} = T_1 V_A^{\gamma-1}$$

$$T_1 T_2 T_3 (V_B V_D V_F)^{\gamma-1} = T_1 T_2 T_3 (V_C V_E V_A)^{\gamma-1}$$

$$\frac{V_B}{V_A} \times \frac{V_D}{V_C} = \frac{V_E}{V_F}$$

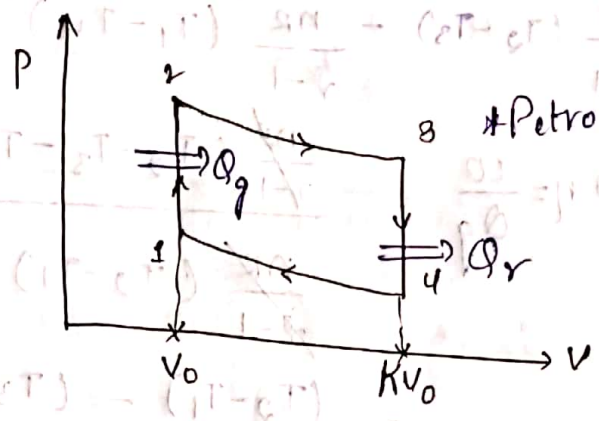
$$\frac{V_E}{V_F} = k^r$$

$$\Rightarrow \eta = \frac{nRT_1 \ln k + nRT_2 \ln k - 2nRT_3 \ln k}{nRT_1 \ln k + nRT_2 \ln k}$$

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$$\eta = \frac{T_1 + T_2 - 2T_3}{T_1 + T_2}$$

Ex! Find out the efficiency of a cycle consists 2 isochoric & 2 adiabatic lines. The volume of the ideal gas changes by K-times within the cycle.



Petrol Engine

Sol!

Along 12!

V = const.

$W_{12} = 0$

$\Rightarrow P \propto T$

$$Q_g = n C_v (T_2 - T_1)$$

$$= n\left(\frac{R}{\gamma - 1}\right)(T_2 - T_1)$$

Along 23!

$Q = 0$

$$W_{23} = \frac{nR}{\gamma - 1}(T_2 - T_3)$$

Along 34!

V = const

$W_{34} = 0$

$P \propto T$

& heat is rejected by the gas.

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along 41 :

$Q = 0$

$$W_{41} = \frac{nR}{\gamma - 1}(T_4 - T_1)$$

$$W_{net} = \cancel{W_{12}}^{0} + W_{23} + \cancel{W_{34}}^{0} + W_{41}$$

$$= \frac{nR}{\gamma - 1}(T_2 - T_3) - \frac{nR}{\gamma - 1}(T_1 - T_4)$$

efficiency $\Rightarrow \eta = \frac{W}{Q_g} = \frac{\frac{nR}{\gamma - 1}(T_2 - T_3 - T_1 + T_4)}{\frac{nR}{\gamma - 1}(T_2 - T_1)}$

$$= \frac{(T_2 - T_1) - (T_3 - T_4)}{T_2 - T_1}$$

$$\boxed{\eta = 1 - \frac{T_3 - T_4}{T_2 - T_1}}$$

(or)

$$V_0 T_2^{\frac{1}{\gamma - 1}} = (kV_0) T_3^{\frac{1}{\gamma - 1}}$$

$$T_2^{\frac{1}{\gamma - 1}} = k T_3^{\frac{1}{\gamma - 1}} \quad - (1)$$

$$V_0 T_1^{\frac{1}{\gamma - 1}} = k V_0 T_4^{\frac{1}{\gamma - 1}}$$

$$T_1^{\frac{1}{\gamma - 1}} = k T_4^{\frac{1}{\gamma - 1}} \quad - (2)$$

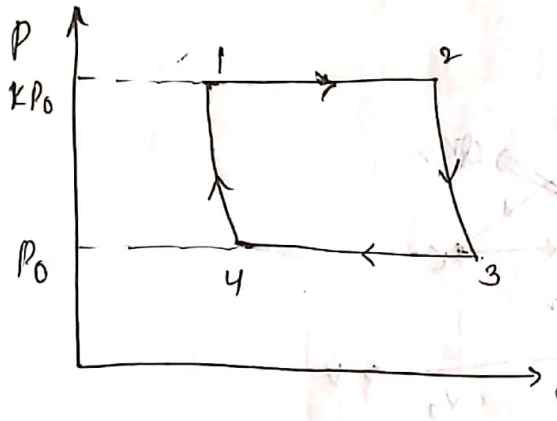
from (1) & (2)

$$\frac{T_3 - T_4}{T_2 - T_1} = k^{\frac{1}{\gamma - 1}}$$

$$\boxed{\eta = 1 - \frac{1}{k^{\gamma - 1}}}$$

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Ex: Find out η of cycle consisting two isobaric and two adiabatic lies. The pressure changes k times with in the cycle & γ given.



Ans: $\eta = 1 - \frac{T_3 - T_4}{T_2 - T_1}$

$\eta = 1 - \frac{1}{k^{\gamma-1/\gamma}}$

Sol:

$$\eta = 1 - \frac{T_3 - T_4}{T_2 - T_1}$$ (from last question)

(previous)

$$T^{\gamma} p^{1-\gamma} = \text{const}$$

$$T p^{\frac{1-\gamma}{\gamma}} = \text{const}$$

$$T_1 (kp_0)^{\frac{1-\gamma}{\gamma}} = T_4 p_0^{\frac{1-\gamma}{\gamma}}$$

$$T_1 k^{\frac{1-\gamma}{\gamma}} = T_4 \quad -(1)$$

$$T_2 (kp_0)^{\frac{1-\gamma}{\gamma}} = T_3 p_0^{\frac{1-\gamma}{\gamma}}$$

$$T_2 k^{\frac{1-\gamma}{\gamma}} = T_3 \quad -(2)$$

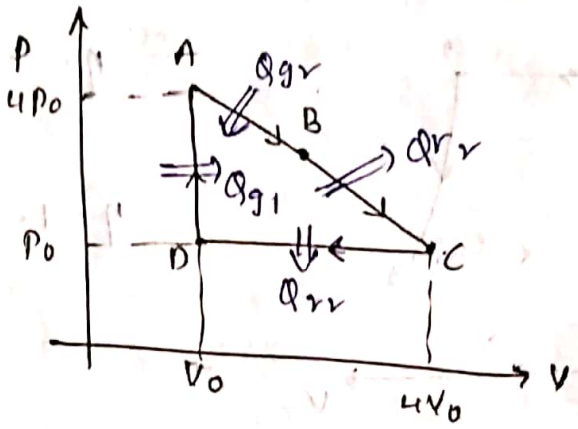
(2) - (1) $\Rightarrow$ $T_3 - T_4 = k^{\frac{1-\gamma}{\gamma}} (T_2 - T_1)$

$$\frac{T_3 - T_4}{T_2 - T_1} = k^{\frac{1-\gamma}{\gamma}}$$

$$\boxed{\eta = 1 - \frac{1}{k^{\frac{\gamma-1}{\gamma}}}}$$

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Ex: During ABC, heat is absorbed by the system till the pt B & after B system rejects heat. Find η of diatomic gas in cycle.



Soln W_{net} = area $= \frac{1}{2}(3V_0)(3P_0)$

$= \frac{9}{2}P_0V_0$

Along DA:

$V = \text{const} \Rightarrow P \propto T$

$Q_{g_1} = nC_V(T_A - T_D)$

$= n\left(\frac{5R}{2}\right)\left(\frac{3P_0V_0}{nR}\right)$

$$\boxed{Q_{g_1} = \frac{15}{2}P_0V_0}$$

Along ABC:

$$\frac{P_0 - 4P_0}{4V_0 - V_0} = \left(\frac{P_0 - P_B}{4V_0 - V_B}\right) = \left(\frac{4P_0 - P_B}{V_0 - V_B}\right)$$

$$\Rightarrow \frac{P_B}{P_0} + \frac{V_B}{V_0} = 5$$

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* $dQ = 0$

$nC_V dT + P dV = 0$

$n\left(\frac{5R}{2}\right) + P\frac{dV}{dT} = 0$

$\frac{5nR}{2} + \left(5P_0 - \frac{2P_0V}{V_0}\right) \times \frac{nR}{\left(5P_0 - \frac{2P_0V}{V_0}\right)} = 0$

$\frac{5}{2} + \frac{(5 - V/V_0)}{(5 - 2V/V_0)} = 0$

$\frac{5 - V/V_0}{5 - 2V/V_0} = -\frac{5}{2}$

$10 - 2V/V_0 = -25 + 10V/V_0$

$35 = 12V/V_0$

$\frac{V}{V_0} = \frac{35}{12}$

$\boxed{V = \frac{35V_0}{12}}$ — put in eqn ①

$P = -\frac{P_0}{V_0}V + c$

$4P_0 = -\frac{P_0}{V_0} \times V_0 + c$

$\boxed{c = 5P_0}$

$\boxed{P = -\frac{P_0V}{V_0} + 5P_0}$ — ①

$PV = nRT$

$\left(\frac{-2P_0V}{V_0} + 5P_0\right) dV = nR\,dT$

$\frac{dV}{dT} = \frac{nR}{\left(5P_0 - \frac{2P_0V}{V_0}\right)}$

$\Rightarrow P = \frac{25P_0}{12}$

$T_B = \frac{P_BV_B}{nR} = \frac{\frac{25P_0}{12} \times \frac{35V_0}{12}}{nR} = \frac{875P_0V_0}{144nR}$

Along AB,

$dQ = dU + dW$

$Q_{g_2} = nC_V dT + P\,dV$

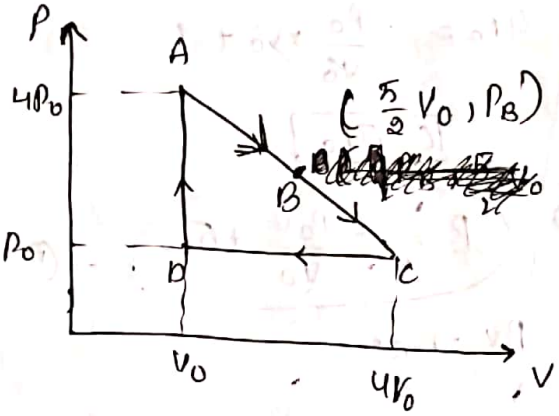
$= (n)\left(\frac{5R}{2}\right)(T_B - T_A) + \frac{1}{2}(4P_0 + P_B)(V_B - V_0)$

$= \frac{5nR}{2}\left(\frac{875P_0V_0}{144nR} - \frac{4P_0V_0}{nR}\right) + \frac{1}{2}\left(\frac{73P_0}{12}\right)\left(\frac{23V_0}{12}\right)$

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$$\eta = \frac{W}{Q_{g_1} + Q_{g_2}}$$

Ex:



$$P_B = \frac{5}{2}P_0$$

net work done, $W = \frac{9}{2}P_0V_0$

Along DA: $V = \text{const} \Rightarrow W = 0$

$$Q_{g_1} = nC_V(T_A - T_D)$$

$$Q_{g_1} = \frac{15}{2}P_0V_0$$

Along AB:

$$Q_{g_2} = nC_V(T_B - T_A) + \frac{1}{2}(4P_0 + P_B)(V_B - V_0)$$

$$= \frac{1}{1} \times \frac{5}{2}\left(\frac{25P_0V_0}{4R} - \frac{4P_0V_0}{R}\right) + \frac{1}{2}\left(\frac{13P_0}{2}\right)\left(\frac{3}{2}V_0\right)$$

$$= \frac{5}{2}\left(\frac{9P_0V_0}{4}\right) + \frac{1}{2}\left(\frac{39P_0V_0}{4}\right)$$

$$= \frac{(45+39)P_0V_0}{8}$$

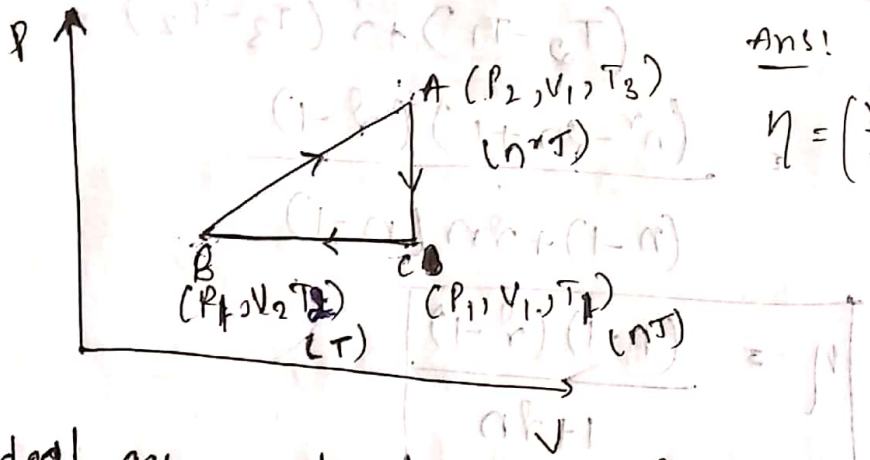
$$Q_{g_2} = \frac{84P_0V_0}{8}$$

$$\eta = \frac{W}{Q_{g_1} + Q_{g_2}} = \frac{\frac{9}{2}P_0V_0}{\frac{(60+84)P_0V_0}{8}} = \frac{9 \times 4 P_0V_0}{144 P_0V_0} = \frac{1}{4}$$

$$\boxed{\eta = 25\%}$$

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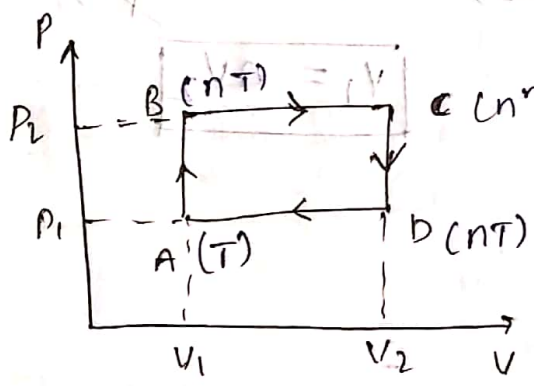
Ex: 1) An ideal gas with an adiabatic exponent γ goes a cycle within which the absolute temp varies n fold. Find the efficiency of the cycle.



Ans!

$$\eta = \left(\frac{\gamma-1}{\gamma+1}\right)\left(\frac{n-1}{n+1}\right)$$

2) An ideal gas whose adiabatic exponent is γ goes through a cycle consisting of 2 isochoric and 2 ~~adiabatic~~ isobaric lines. Find the efficiency of cycle if the absolute temp of gas rises n-times both in isochoric heating & isobaric expansion.



Ans!

$$\eta = \frac{(\gamma-1)(n-1)}{1+n\gamma}$$

Sol:

2)

$$W = (V_2 - V_1)(P_2 - P_1)$$

$$= V_2P_2 - V_2P_1 - V_1P_2 + P_1V_1$$

$$= nRT_3 - nRT_4 + nRT_2 + nRT_1$$

$T_A = T$

$T_B = nT$

$T_D = nT$

$T_C = n^2T$

$$Q_1 = \frac{R}{\gamma-1}(T_2 - T_1)$$

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$$Q_2 = \frac{\gamma R}{\gamma - 1}(T_3 - T_2)$$

$$\eta = \frac{(T_3 - T_1 - T_4 - T_2)(\gamma - 1)}{(T_2 - T_1) + \gamma(T_3 - T_2)}$$

$$= \frac{(n^2 - 2n + 1)(\gamma - 1)}{(n-1) + \gamma n(n-1)}$$

$$\boxed{\eta = \frac{(n-1)(\gamma - 1)}{1 + \gamma n}}$$

1) $$W = \frac{1}{2}(P_2 - P_1)(-V_2 + V_1) \quad -(1)$$

BC: $p = \text{const}$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow V_1 = V_2\left(\frac{nT}{T}\right)$$

$$\boxed{V_1 = nV_2}$$

AC: $V = \text{const}$

$$\frac{P_1}{T_1} = \frac{P_2}{T_3}$$

$$\Rightarrow \frac{P_1}{T} = \frac{P_2}{nT} \Rightarrow \boxed{P_2 = nP_1}$$

(1) $\Rightarrow$ $$W = \frac{1}{2}(n-1)(1+n)P_1V_2$$

heat absorbed in process AB.

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$$Q_g = nC\Delta T$$

$$= \frac{nR}{2}\left(\frac{\gamma+\gamma-1}{\gamma-1}\right)\left(\frac{P_2V_1}{nR} - \frac{P_1V_2}{nR}\right)$$

$$C = C_V + \frac{R}{2}$$

$$= \frac{R}{\gamma-1} + \frac{R}{2}$$

$$= \frac{1}{2}\left(\frac{\gamma+1}{\gamma-1}\right)(P_2V_1 - P_1V_2)$$

$$= \frac{1}{2}\left(\frac{\gamma+1}{\gamma-1}\right)(n^{\gamma}P_1V_2 - P_1V_2)$$

$$= \frac{1}{2}\left(\frac{\gamma+1}{\gamma-1}\right)(n^{\gamma}-1)(P_1V_2)$$

$$\eta = \frac{W}{Q} = \frac{\frac{1}{2}(n-1)(n+1)(n-1)P_1V_2}{\frac{1}{2}\left(\frac{\gamma+1}{\gamma-1}\right)(n+1)(n-1)P_1V_2}$$

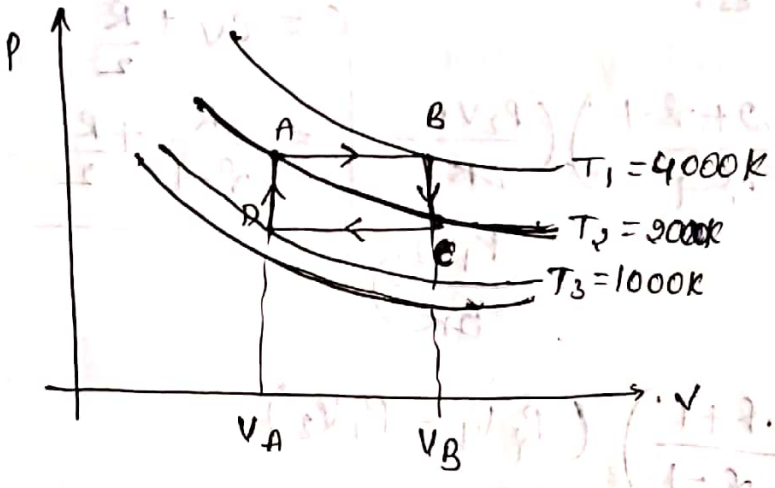
$$= \left(\frac{\gamma-1}{\gamma+1}\right)\left(\frac{-1+n}{n+1}\right)$$

$$\boxed{\eta = \left(\frac{\gamma-1}{\gamma+1}\right)\left(\frac{n-1}{n+1}\right)}$$

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05/11/24 Tuesday

Ex 1

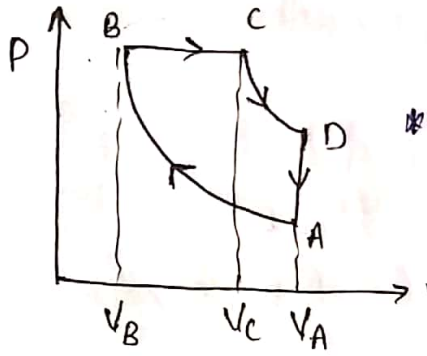


when 1 mole of an ideal mono-atomic gas is taken through the paths AB, BC, CD & DA. Calculate the change in internal energy ΔU, workdone by the gas W & heat absorbed by the gas Q along each path. Also find η of the complete cycle ABCDA. Given that $V_A = 1m^3$, $V_B = 2m^3$, $R = 8.3\ J/mol/K$

Ans: $\eta = 15.38\%$

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Ex: 1mole of diatomic ideal gas ($\gamma = 1.4$) is taken through a cyclic process starting from pt 'A'. Here process AB is adiabatic compression, BC is isobaric, CD is adiabatic expansion, DA is isobaric. The volume ratios are $\frac{V_A}{V_B} = 16$, $\frac{V_C}{V_B} = 2$ & $T_A = 300K$. Find Temp of gas at pts B & D. Also calculate efficiency of the cycle.



* diesel engine *

$V_A = V_D$

Ans: $T_B = 909K$

$T_D = 791K$

$\eta = 61.4\%$

Sol:

~~$W_{AB} = \frac{nR}{\gamma - 1}(T_A - T_B)$~~

AB: $T V^{\gamma-1} = \text{const}$

$T_A V_A^{\gamma-1} = T_B V_B^{\gamma-1}$

$T_A \times \left(\frac{V_A}{V_B}\right)^{1.4-1} = T_B$

$300 \times (16)^{0.4} = T_B$

$300 \times (16)^{2/5} = T_B$

$300 \times (16 \times 16)^{1/5} = T_B$

$300 \times 2\,(8)^{1/5} = T_B$

$600 \times 8^{1/5} = T_B$

$T_B = 600 \times 2^{3/5}$

$\boxed{T_B = 909.43K}$

BC: $\frac{V_B}{T_B} = \frac{V_C}{T_C}$

$T_C = T_B \left(\frac{V_C}{V_B}\right)$

$\boxed{T_C = 1818K}$

CD: $T V^{\gamma-1} = \text{const}$

$T_C V_C^{\gamma-1} = T_D V_D^{\gamma-1}$

$T_C \left(\frac{V_C}{V_D}\right)^{\gamma-1} = T_D$

$1818 \left(\frac{V_C}{V_A}\right)^{\gamma-1} = T_D$

$1818 \left(\frac{2}{16}\right)^{0.4} = T_D$

$\boxed{T_D = 791.3K}$

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$$Q_{AB} = \frac{nR}{\gamma - 1}(T_A - T_B)$$

$$= \frac{1 \times R}{1.4 - 1}(300 - 909)$$

$$\boxed{Q_{AB} = 1522.5R}$$

$\boxed{Q_{AB} = 0}$ → adiabatic

$$Q_{BC} = nC_p\Delta T$$

$$= 1 \times \frac{7R}{2}(909)$$

$$\boxed{Q_{BC} = 3181.5R}$$

$$\Rightarrow Q_g = 0 + 3181.5R = \underline{3181.5R}$$

$\boxed{Q_{CD} = 0}$ → adiabatic

$$Q_{DA} = nC_v\Delta T$$

$$= 1 \times \frac{5R}{2} \times (491 - 300)$$

$$= \frac{5R}{2} \times 491$$

$$\boxed{Q_{DA} = 1227.5R}$$

$$Q_r = 0 + 1227.5R = \underline{1227.5R}$$

$$\eta = 1 - \frac{Q_r}{Q_g}$$

$$= 1 - \frac{1227.5R}{3181.5R}$$

$$= \frac{1954}{3181.5} = 0.6142 \Rightarrow \boxed{\eta = 61.4\%}$$

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Heat Engine :

* It is a device that converts "heat" into "mechanical work."
* It has mainly three parts.

① Source :

* It is a hot body maintained at very high temp (T_1 K).
* It's thermal capacity is ∞. $\left(\frac{\Delta Q}{\Delta T} = \infty\right)$

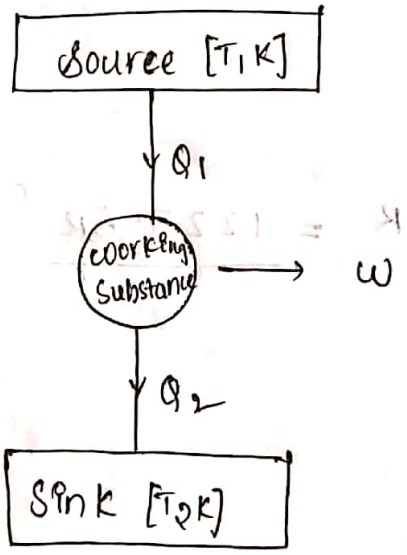
② Sink :

* It is a cold body maintained at very low temp (T_2 K).
* It's thermal capacity is ∞.

③ Working Substance :

* An ideal gas taken in a container with a movable piston is called working substance.

* Block diagram of heat engine



From energy conservation,

$$Q_1 = Q_2 + W$$

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$$W = Q_1 - Q_2$$

efficiency of any heat engine,

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$$\boxed{\eta = 1 - \frac{Q_2}{Q_1}} \quad *$$

$$* \quad \eta = 1 - \frac{|Q_r|}{Q_{ab}}$$

* For a carnot reversible engine,

$$\eta = 1 - \frac{T_2}{T_1} \quad \text{--- (1)}$$

$$\eta = 1 - \frac{Q_2}{Q_1} \quad \text{--- (2)}$$

from (1) & (2) $\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$

$$\Rightarrow \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \Rightarrow \boxed{\frac{Q_1}{T_1} = \frac{Q_2}{T_2} = - - -}$$

— for any no. of sources & sinks

$$\therefore \quad ** \quad \boxed{\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}} \quad **$$

* generally, $\eta < 1$.

* for ideal heat engine, $\eta = 1$

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Imp points!

1) $\eta = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$

η can be increased by

a) increasing the temp diff i.e, $(T_1 - T_2) \uparrow$

b) increasing the source temp only
i.e, $\xi, T_1 (\uparrow)$

c) decreasing the sink temp only

* decreasing T_2 is more effective than increasing T_1. i.e, $T_2 (\downarrow)$

Ex!

$$\eta = 1 - \frac{100}{400} = 1 - \frac{1}{4} = 3/4 = 75\%$$

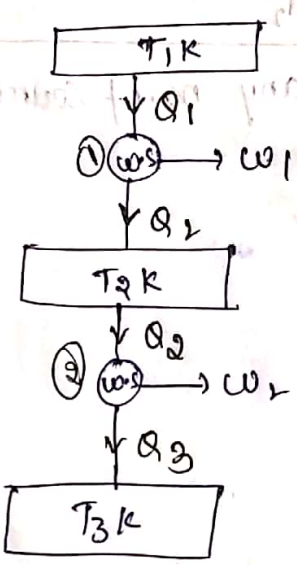
now, $\eta_1 = 1 - \frac{90}{400} = \frac{31}{40} = 77.5\%$

(only sink temp decreases by 10K

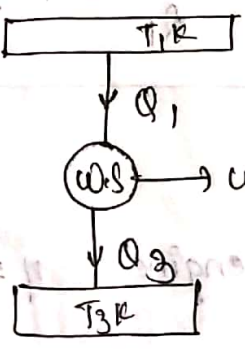
$\eta_2 = 1 - \frac{100}{410} = \frac{31}{41} = 75.6\%$

(only source temp increases by 10K)

2)



when engines are series



$\eta = 1 - \frac{T_3}{T_1}$

$\frac{T_3}{T_1} = 1 - \eta$

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$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} = \frac{Q_3}{T_3} \quad -①$$

$$\eta_1 = \frac{w_1}{Q_1} = 1 - \frac{T_2}{T_1} \quad -②$$

∴ out put of engine,

$$w_1 = Q_1\left(1 - \frac{T_2}{T_1}\right)$$

& $$\eta_2 = \frac{w_2}{Q_2} = 1 - \frac{T_3}{T_2} \quad -③$$

∴ output of 2nd engine,

$$w_2 = Q_2\left(1 - \frac{T_3}{T_2}\right)$$

we have,

$$\eta_1 = 1 - \frac{T_2}{T_1} \Rightarrow \frac{T_2}{T_1} = 1 - \eta_1$$

$$\eta_2 = 1 - \frac{T_3}{T_2} \Rightarrow \frac{T_3}{T_2} = 1 - \eta_2$$

now, $$(1-\eta_1)(1-\eta_2) = \frac{T_2}{T_1} \times \frac{T_3}{T_2}$$

$$(1-\eta_1)(1-\eta_2) = \frac{T_3}{T_1}$$

$$(1-\eta) = (1-\eta_1)(1-\eta_2)$$

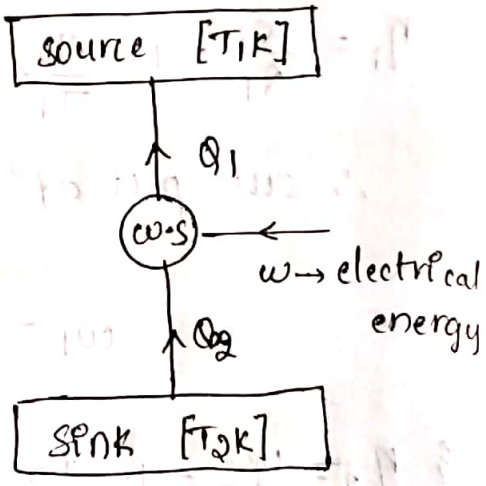
$$1-\eta = 1 + \eta_1\eta_2 - \eta_1 - \eta_2$$

** $$\boxed{\eta = \eta_1 + \eta_2 - \eta_1\eta_2}$$ //

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Refrigirator (Heat Pump):

* Refrigirator is a heat engine working in reverse direction.



* from energy conservation,

$$Q_2 + W = Q_1$$

$$\boxed{W = Q_1 - Q_2}$$

* **Coefficient of performance:**

$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}$$

** $$\boxed{\beta = \frac{1}{\frac{Q_1}{Q_2} - 1} = \frac{1}{\frac{T_1}{T_2} - 1} = \frac{T_2}{T_1 - T_2} = \frac{Q_2}{W}}$$

generally, $\beta > 1$.

Relation b/w η & β:

$$\eta = \frac{T_1 - T_2}{T_1} \Rightarrow \eta = 1 - \frac{T_2}{T_1}$$

$$\Rightarrow \frac{T_2}{T_1} = 1 - \eta$$

$$\beta = \frac{T_2}{T_1 - T_2}$$

$$\eta\beta = \frac{T_2}{T_1}$$

$$\eta\beta = 1 - \eta$$

** $$\boxed{\beta = \frac{1-\eta}{\eta}}$$ (or) $$\boxed{\eta = \frac{1}{1+\beta}}$$

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Ex1 A carnot engine is having an efficiency of 10% is being used as a refrigirator. If workdone on the refrigirator is 10 J. Then calculate the amount of heat absorbed from the reservior at lower temp.

$$\eta = 10\% = \frac{1}{10}$$

$$\beta + 1 = 10$$

$$\boxed{\beta = 9}$$

$$q = \frac{Q_2}{W}$$

$$Q_2 = qW = 90$$

$$\boxed{Q_2 = 90\ J}$$

& heat given to source,

$$Q_1 = Q_2 + W$$

$$= 90 + 10$$

$$\boxed{Q_1 = 100\ J}$$

Ex! How much energy in watt hour may be required to convert 2 kg of water into ice at 0°C. Assuming that the refrigirator is ideal. Given temp of freezer is -15°C and room temp is 25°C & initial temp of water is 25°C.

$$T_1 = 25 + 273 = 298$$

$$T_2 = 273 - 15 = 258$$

$$S_w = 1\ cal/gm\ ^\circ C = 4200\ J/kg^\circ c$$

$$L_{ice} = 80\ cal/gm = 3.36 \times 10^5\ J/kg$$

heat required to convert wate into ice at 0°C

$$Q_2 = mS_w\Delta T + mL_{ice}$$

$$= 2 \times 4200 \times 25 + 2 \times 3.36 \times 10^5$$

$$\boxed{Q_2 = 8.82 \times 10^5\ J}$$

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heat reject to surrounding,

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$Q_1 = Q_2\left(\frac{T_1}{T_2}\right)$$

$$= 8.82 \times 10^5 \times \frac{298}{258}$$

$$\boxed{Q_1 = 10.15 \times 10^5 \text{ J}}$$

Energy required = w

$$w = Q_1 - Q_2$$

$$= 10.15 \times 10^5 - 8.82 \times 10^5$$

$$w = 1.33 \times 10^5 \text{ J}$$

$$w = \frac{1.33 \times 10^5 \times 3}{60 \times 60} \text{ whr}$$

$$= \frac{1330}{36} \text{ whr}$$

$$\boxed{w = 36.96 \text{ whr}}$$

Ex: 2 carnot engines A & B are operated in series the 1st engine A recieve heat at 800K & rejects to a reservior at T_2 K. The second engine B recieves the heat rejected by 1st engine & intern rejects to a heat reservior at 300K. Calculate the temp T_2 for the following cases.

1) when out puts of 2 engines are equal

2) when efficiency of 2 engines are equal.

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1) $\frac{Q_1}{T_1} = \frac{Q_2}{T_2} = \frac{Q_3}{T_3}$ 2) $\eta_1 = \eta_1$

$\frac{Q_1}{800} = \frac{Q_2}{T_2}$ $\frac{T_2}{800} = \frac{300}{T_2}$

$\frac{W_1}{Q_1} = 1 - \frac{T_2}{T_1}$, $\frac{W_2}{Q_2} = 1 - \frac{T_3}{T_2}$ $T_2^2 = 240 \times 100$

$T_2 = 2\sqrt{6} \times 100$

$W_1 = W_2$ $Q_1\left(1 - \frac{T_2}{T_1}\right) = Q_2\left(1 - \frac{T_3}{T_2}\right)$ $T_2 = 490\,K$

$\frac{800 Q_2}{T_2}\left(1 - \frac{T_2}{T_1}\right) = Q_2\left(1 - \frac{T_3}{T_2}\right)$

$\Rightarrow$ $T_2 = 550\,K$

Reversible process:

A process which can be retraced back in the opposite direction in such a way that the system passes through the same states as in direct process. & finally system & surrounding aquires the initial condns.

Condns for the process to be reversible:

* The change must takes place at a very slow rate (quasi-static process).
* There should be no loss of energy due to conduction, convection, friction, viscosity etc.....
* The system must always be in thermal & chemical eqm with the surroundings.

Examples:

1. Fusion of ice, vapourisation of water. – reversible
2. gradual compression & extension of spring – reversible

Irreversible process:

The process which cannot be retraced back in opposite direction is called irreversible process. Almost all process

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of nature are irreversible.

Examples:

1. work done against friction.
2. diffusion of gases into one another.
3. Heat produced due to flow of current through wire.
4. Rainfall.

Limitations of 1st law of Thermodynamics:

1. It doesnot tell about the direction of flow of heat.
2. It doesnot tell about in which condns heat cannot be converted into work.

Ex:

* A revolving wheel stops due to friction.

According to first law, K.E of wheel is converted into heat. But it fails to explain why heat energy can't be converted into K.E of wheel.

3. It doesnot tell about the amount of heat that can be converted into work.

* Second law of thermodynamics provides answers for these questions.

Second Law of Thermodynamics:

* Kelvin-planck Statement - It is impossible to ~~Convert~~ construct an engine that can convert heat completly into work without producing any other effect.

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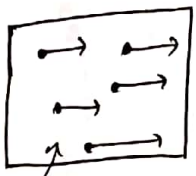
* There will be no machine that has 100% efficiency.

* Clausius statement – For any self acting machine, it is impossible to transfer heat from a colder body to a hoter body without the aid of external agency.

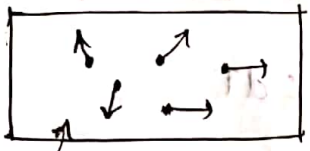
Entropy:

06|11|24| wednesday

* It tells us the disorder (or) randomness of system.



entropy is less



entropy is more

* Entropy is a state dependent quantity but does not depends on path followed.

* when the system changes from initial state to final state, change in entropy is

$$ds = \frac{dQ}{T} = \frac{\text{heat absorbed / rejected by system}}{\text{absolute temp.}}$$

* Total change in entropy,

$$\Delta S = \int ds = \int_i^f \frac{dQ}{T} \quad \text{unit: J/K}$$

Points:

* If heat is transfered at const temp. Then,

$$\Delta S = \int \frac{dQ}{T} = \frac{1}{T}\int dQ \Rightarrow \Delta S = \frac{\Delta Q}{T}$$

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For ex,

when ice melts into water at 0°c,

$$\Delta S = \frac{mL_{ice}}{T}$$

$T = 0 + 273 = 273K$

here $\Delta S = +ve$, i.e, entropy increases during melting of ice into water at 0°c.

* During the change of temp,

$$dQ = mc\,dT$$

$$\Delta S = \int_{T_i}^{T_f} \frac{dQ}{T}$$

$$\Delta S = mc \int_{T_i}^{T_f} \frac{dT}{T}$$

$$\boxed{\Delta S = mc \ln (T_f / T_i)}$$

Ex: Find change in entropy of 1kg of ice at 0°c is converted into water at 100°c.

Sol:

$$\Delta S = \int \frac{dQ}{T}$$

$$= \int \frac{(dm)L + mc\,dT}{T \quad T}$$

$$\Delta S = \int \frac{dm}{T} L + \int_{T_i}^{T_f} mc_w \frac{dT}{T}$$

$$\Delta S = \frac{L}{T} m + m c_w \ln (T_f / T_i)$$

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$$\Delta S = \frac{m L_{ice}}{T} + m c_w \ln\left(\frac{T_f}{T_i}\right)$$

here, $m = 1kg$

$T_i = T = 273K$

$L_{ice} = 80\ cal/gm = 3.36 \times 10^5 J/kg$

$C_w = 1\ cal/gm\,^oc = 4200\ J/kg\,k$

$T_f = 100 + 273 = 373k$

$$\Delta S = \frac{1 \times 3.36 \times 10^5}{273} + 1 \times 4200 \ln\left(\frac{373}{273}\right)$$

$\Delta S =$

In Case of gases:

① For Isochoric process:

$$dQ = nC_v dT$$

$$\Delta S = \int \frac{dQ}{T} = \int_{T_i}^{T_f} \frac{nC_v dT}{T} = nC_v \ln(T_f/T_i)$$

② For Isobaric process:

$$dQ = nC_p dT$$

$$\Delta S = \int \frac{dQ}{T} = \int_{T_i}^{T_f} \frac{nC_p dT}{T} = nC_p \ln(T_f/T_i)$$

③ For adiabatic process:

$dQ = 0 \Rightarrow \boxed{dS = 0}$

change in entropy = 0 i.e, $\boxed{S = const}$

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Ex! Find change in entropy of an ideal gas during change of state from (V_i, T_i) to (V_f, T_f) in undefined process.

Sol! From 1st law of thermodynamics,

$$dQ = dU + dW$$

$$dQ = nC_v\,dT + P\,dV$$

now,

$$\Delta S = \int_{T_i}^{T_f} \frac{dQ}{T}$$

$$= \int_{T_i}^{T_f} \frac{nC_v\,dT}{T} + \int_{V_i}^{V_f} \frac{P\,dV}{T} \qquad [\because PV = nRT]$$

$$\frac{P}{T} = \frac{nR}{V}$$

$$\Delta S = nC_v \ln(T_f/T_i) + \int_{V_i}^{V_f} nR\,\frac{dV}{V}$$

$$\boxed{\Delta S = nC_v \ln(T_f/T_i) + nR \ln(V_f/V_i)}$$

Point!

① For isothermal process:

$T = \text{const}$

$$\frac{T_f}{T_i} = 1$$

$$\ln(T_f/T_i) = 0$$

$$\boxed{\Delta S = nR \ln(V_f/V_i)}$$

② For isobaric process:

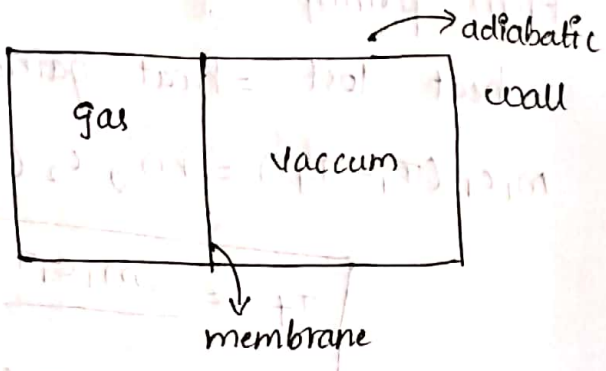
$V = \text{const} \Rightarrow \ln\left(\frac{V_f}{V_i}\right) = 0$

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$$\Delta S = nR \ln (T_f/T_i)$$

Adiabatic free Expansion:

when membrane is punched, gas rushes into vaccum and hence randomness is increased.



So entropy of gas increases.

here $dQ = 0$

$dW = 0$

$\Rightarrow \boxed{dU = 0}$

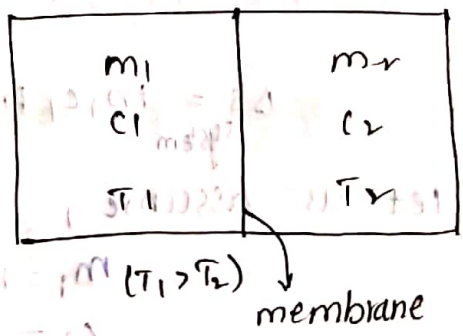
$\Rightarrow \boxed{T = \text{const}}$

$\therefore$ change in entropy,

$$\boxed{\Delta S = nR \ln (V_f/V_i)}$$

* This is the special case where $dQ = 0$ but $ds \neq 0$.
* It is irreversible process.

Ex: Two liquids are taken in the insulated chamber seperated by membrane as shown. when the membrane is punchered. liquids allowed to mix. Find change in entropy of the system.



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Sol:

Let T_f – final temp of mixture

From principle of mixtures,

heat lost = heat gained.

$$m_1 c_1 (T_1 - T_f) = m_2 c_2 (T_f - T_2)$$

$$T_f = \frac{m_1 c_1 T_1 + m_2 c_2 T_2}{m_1 c_1 + m_2 c_2}$$

here $\boxed{T_1 > T_f > T_2}$

change in entropy of first liquid,

$$\Delta S_1 = m_1 c_1 \ln (T_f / T_1) \quad (T_f < T_1)$$

clearly $\Delta S_1 = -ve$

i.e, entropy decreases.

change in entropy of 2nd liquid,

$$\Delta S_2 = m_2 c_2 \ln (T_f / T_2) \quad (T_f > T_2)$$

clearly $\Delta S_2 = +ve$

i.e, entropy increases.

Total change in entropy of system,

$$\Delta S = \Delta S_1 + \Delta S_2$$

$$\Delta S_{system} = m_1 c_1 \ln (T_f / T_1) + m_2 c_2 \ln (T_f / T_2)$$

let us assume,

$$m_1 = m_2 = m$$

$$c_1 = c_2 = c \quad \Rightarrow T_f = \frac{T_1 + T_2}{2}$$

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$$\Delta S_{system} = mc\left[\ln(T_f/T_1) + \ln(T_f/T_2)\right]$$

$$= mc\left[\ln(T_f^2/T_1T_2)\right]$$

$$\Delta S_{system} = mc\ln\left(\frac{\left(\frac{T_1+T_2}{2}\right)^2}{T_1T_2}\right)$$

▲ we have,

$AM \geq GM$

$AM = GM$ when $T_1 = T_2$

but here, $\boxed{T_1 > T_2}$

$\therefore AM > GM$

$$\frac{T_1+T_2}{2} > \sqrt{T_1T_2}$$

$$\left(\frac{T_1+T_2}{2}\right)^2 > T_1T_2$$

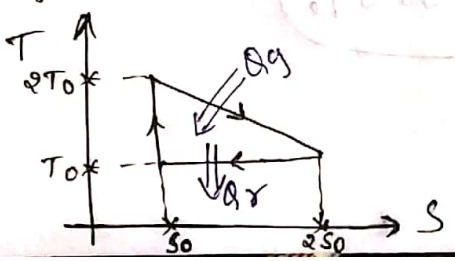
$$\Rightarrow \boxed{\Delta S_{system} = +ve}$$

i.e, entropy of system increases.

Note:

entropy of system increases always in case of irreversible process.

Ex: Find efficiency of reversible heat engine cycle.



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Soll

we have,

$$ds = \frac{dQ}{T}$$

$$dQ = T\,ds$$

$$\boxed{Q = \int dQ = \int T \cdot ds = \text{area of } T\text{-}s \text{ graph}}$$

If $ds = +ve \Rightarrow Q = +ve$ i.e, heat absorbed by system.

$ds = -ve \Rightarrow Q = -ve$ i.e, heat rejected by system

$ds = 0 \Rightarrow Q = 0$

now,

$$Q_g = \frac{1}{2}(2T_0 + T_0)(2s_0 - s_0)$$

$$Q_g = \frac{3}{2} T_0 s_0$$

$$Q_r = T_0 s_0$$

now, $$\eta = 1 - \frac{Q_r}{Q_g}$$

$$= 1 - \frac{T_0 s_0}{\frac{3}{2} T_0 s_0}$$

$$= 1 - \frac{2}{3}$$

$$\boxed{\eta = 1/3}$$

$$\eta = 33.3\%$$

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Ex: An inventor claims to have developed an engine that during a certain time interval takes in 110 MJ of heat at 415K rejects 50MJ at 212K while manages to do 16.7 KWhr of work. Do you agree with the inventors claim?

$$\eta_c = \frac{w}{Q_{ab}} = \frac{16.7 \times 36 \times 10^3}{110 \times 10^6} = \frac{167 \times 36}{11 \times 10^3} \times 10^{-3}$$

$$\boxed{\eta_c = 55\%}$$

Maximum theoritical efficiency,

$$\eta_t = 1 - \frac{T_2}{T_1} = 1 - \frac{212}{415} = \frac{203}{415}$$

$$\boxed{\eta_t = 49\%}$$

we donot agree with his claim.

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